ECE120: Basic Electronics Engineering Workshop

Comprehensive University Question Bank, Viva-Voce & Laboratory Problem Suite

LPU Department of Electronics & Communication Engineering
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Chapter 1

Unit 1: Workshop Instruments, Components, Trainer Module & Display Devices

1.1 Section A: Multiple Choice Practice Questions (60 Curated Problems)

The following 60 multiple-choice questions reflect the official workshop evaluation bank, categorized by fundamental learning outcome and laboratory testing difficulty.

Q1. Which component is commonly used to limit the flow of electric current? [Tag: Familiarization of electronic components | Easy]

(a) Resistor
(b) Transistor
(c) Diode
(d) Capacitor

Q2. Which component stores electrical energy in an electric field? [Tag: Familiarization of electronic components | Easy]

(a) Capacitor
(b) Diode
(c) Fuse
(d) Resistor

Q3. What is the main function of a diode in a simple circuit? [Tag: Familiarization of electronic components | Easy]

(a) To store current in two directions
(b) To produce sound from light
(c) To increase resistance in all directions
(d) To allow current in one direction

Q4. What does LED stand for? [Tag: Familiarization of electronic components | Easy]

(a) Light Energy Detector
(b) Low Energy Device
(c) Linear Electrical Display
(d) Light Emitting Diode

Q5. Which component is commonly used as an electronic switch or amplifier? [Tag: Familiarization of electronic components | Easy]

(a) Resistor
(b) Inductor
(c) Capacitor

(d) Transistor

Q6. What is the purpose of a fuse in an electronic circuit? [Tag: Familiarization of electronic components | Easy]

(a) To store electrical charge
(b) To display a binary number
(c) To increase the supply voltage
(d) To protect against excessive current

Q7. Which component is designed to store energy in a magnetic field? [Tag: Familiarization of electronic components | Easy]

(a) Inductor
(b) Diode
(c) LED
(d) Resistor

Q8. What is an integrated circuit, commonly called an IC? [Tag: Familiarization of electronic components | Easy]

(a) A group of circuits on one chip
(b) A single wire with insulation
(c) A device used only for measuring voltage
(d) A mechanical switch with many contacts

Q9. What is the main purpose of a digital trainer module? [Tag: Familiarization of the digital trainer module | Easy]

(a) To measure room temperature only
(b) To charge batteries automatically
(c) To convert mechanical motion into sound
(d) To test and build digital circuits

Q10. Which source on a digital trainer is commonly used to provide logic 0 or logic 1 inputs? [Tag: Familiarization of the digital trainer module | Easy]

(a) Transformer core
(b) Heat sink
(c) Speaker
(d) Logic switch

Q11. Which indicator on a digital trainer commonly shows the state of a logic signal? [Tag: Familiarization of the digital trainer module | Easy]

(a) LED indicator
(b) Fuse holder
(c) Power transformer
(d) Variable resistor

Q12. What is the function of a regulated power supply in a digital trainer? [Tag: Familiarization of the digital trainer module | Easy]

(a) To provide a stable DC voltage
(b) To produce mechanical rotation
(c) To store programs permanently
(d) To measure resistance directly

Q13. Which device on a digital trainer can provide regularly repeating digital pulses? [Tag: Familiarization of the digital trainer module | Easy]

(a) Fuse
(b) Potentiometer
(c) Clock generator
(d) Loudspeaker

Q14. What does a common seven-segment display show? [Tag: Familiarization of the digital trainer module | Easy]

(a) Only resistance values
(b) Only analog waveforms
(c) Magnetic field lines
(d) Decimal digits

Q15. Which instrument can generate a sine wave for testing an electronic circuit? [Tag: Familiarization of input sources and display devices related to electronic circuits | Easy]

(a) Function generator
(b) Digital multimeter
(c) Seven-segment display
(d) Loudspeaker

Q16. Which source provides a constant electrical voltage for a circuit? [Tag: Familiarization of input sources and display devices related to electronic circuits | Easy]

(a) Microphone
(b) LED display
(c) DC power supply
(d) Oscilloscope

Q17. Which device converts sound waves into an electrical signal? [Tag: Familiarization of input sources and display devices related to electronic circuits | Easy]

(a) Microphone
(b) Relay
(c) Speaker
(d) Lamp

Q18. Which device converts an electrical signal into sound? [Tag: Familiarization of input sources and display devices related to electronic circuits | Easy]

(a) Photodiode
(b) Microphone
(c) Speaker
(d) Resistor

Q19. Which instrument is used to observe the waveform of an electrical signal? [Tag: Familiarization of input sources and display devices related to electronic circuits | Easy]

(a) Function switch
(b) Soldering iron
(c) Oscilloscope
(d) Battery holder

Q20. Which display device uses liquid crystals to show information? [Tag: Familiarization of input sources and display devices related to electronic circuits | Easy]

(a) LCD
(b) Diode
(c) Transformer
(d) LED

Q21. A resistor with color bands brown, black, and red has an approximate resistance of: [Tag: Familiarization of electronic components | Medium]

(a) 100 Ω
(b) 10 k Ω
(c) 100 k Ω
(d) 1 k Ω

Q22. A circuit requires a 5 V supply and a component must be limited to 10 mA . What resistor value is closest for a simple series connection when the component drop is negligible? [Tag:

Familiarization of electronic components | Medium]

(a) 250 Ω
(b) 5 k Ω
(c) 50 Ω
(d) 500 Ω

Q23. Which component is most suitable for allowing current in one direction while blocking it in the opposite direction? [Tag: Familiarization of electronic components | Medium]

(a) Resistor
(b) Transformer
(c) Diode
(d) Capacitor

Q24. An electrolytic capacitor is connected directly across a DC supply. Which practice is essential? [Tag: Familiarization of electronic components | Medium]

(a) Reverse its marked terminals
(b) Connect it without polarity
(c) Match its polarity correctly
(d) Short its terminals initially

Q25. In a transistor used as an electronic switch, the transistor is intended to operate mainly in which two regions? [Tag: Familiarization of electronic components | Medium]

(a) Leakage and thermal
(b) Active and breakdown
(c) Reverse and avalanche
(d) Cutoff and saturation

Q26. A fuse is placed in series with the supply of an electronic circuit primarily to: [Tag: Familiarization of electronic components | Medium]

(a) Increase the circuit voltage
(b) Protect against excessive current
(c) Limit normal signal frequency
(d) Improve the circuit power factor

Q27. Which component is commonly used to vary resistance manually in order to adjust voltage or signal level? [Tag: Familiarization of electronic components | Medium]

(a) Crystal oscillator
(b) Rectifier diode
(c) Potentiometer
(d) Photodiode

Q28. Before connecting an IC on a digital trainer module, the safest first action is to: [Tag: Familiarization of the digital trainer module | Medium]

(a) Switch off the module supply
(b) Apply the maximum supply voltage
(c) Connect every input to ground
(d) Short the output terminals

Q29. A digital trainer's logic switch is connected to an IC input. What is its main purpose? [Tag: Familiarization of the digital trainer module | Medium]

(a) To measure analog resistance
(b) To provide selectable logic levels
(c) To amplify output current continuously
(d) To display the circuit waveform

Q30. If a trainer module's logic probe indicates a high state at an output, the output voltage is generally: [Tag: Familiarization of the digital trainer module | Medium]

(a) Near the circuit ground level

(b) Near the designated logic supply level
(c) Always equal to the input voltage
(d) Always equal to zero volts

Q31. A seven-segment display on a trainer module shows an incorrect digit even though the logic inputs appear correct. Which check is most relevant first? [Tag: Familiarization of the digital trainer module | Medium]

(a) Replace the display with a resistor
(b) Increase the supply without measuring
(c) Check the segment wiring and common terminal
(d) Disconnect the ground connection

Q32. What is the main function of a clock generator on a digital trainer module? [Tag: Familiarization of the digital trainer module | Medium]

(a) Convert DC directly into sound
(b) Measure the resistance of an IC
(c) Provide periodic digital pulses
(d) Regulate the brightness of all LEDs

Q33. A digital IC input is left unconnected on a trainer module. Why can this cause unreliable operation? [Tag: Familiarization of the digital trainer module | Medium]

(a) The input automatically becomes a regulated output
(b) The input permanently disables the power supply
(c) The input may assume an undefined logic state
(d) The input always produces a precise logic zero

Q34. When testing a logic circuit on a trainer, why should the circuit ground and instrument ground be connected properly? [Tag: Familiarization of the digital trainer module | Medium]

(a) To increase the logic supply frequency
(b) To convert binary signals into analog signals
(c) To prevent all output transitions
(d) To establish a common voltage reference

Q35. A function generator is connected to an amplifier input. Which setting determines how large the applied signal is? [Tag: Familiarization of input sources and display devices related to electronic circuits | Medium]

(a) Time-base scale
(b) Duty-cycle label
(c) Frequency
(d) Amplitude

Q36. A sine wave displayed on an oscilloscope occupies four vertical divisions. If the vertical scale is 0.5 V/div , what is the peak-to-peak voltage? [Tag: Familiarization of input sources and display devices related to electronic circuits | Medium]

(a) 0.5 V
(b) 4.0 V
(c) 2.0 V
(d) 1.0 V

Q37. An oscilloscope shows five complete cycles across ten horizontal divisions. If the time scale is 0.2 ms/div , what is the approximate frequency? [Tag: Familiarization of input sources and display devices related to electronic circuits | Medium]

(a) 100 Hz
(b) 25 kHz
(c) 2.5 kHz
(d) 250 Hz

Q38. Which input device changes its resistance significantly in response to incident light and can

be used in a light-sensing circuit? [Tag: Familiarization of input sources and display devices related to electronic circuits | Medium]

(a) Piezoelectric buzzer
(b) LDR
(c) Thermistor
(d) Relay coil

Q39. A microphone is connected to an electronic circuit as an input source. It primarily converts: [Tag: Familiarization of input sources and display devices related to electronic circuits | Medium]

(a) Sound into an electrical signal
(b) Light into resistance
(c) DC voltage into magnetic flux
(d) Digital data into heat

Q40. Which display device is most appropriate when a circuit must indicate the ON or OFF state of a digital output immediately? [Tag: Familiarization of input sources and display devices related to electronic circuits | Medium]

(a) Thermistor
(b) Transformer
(c) LED
(d) Variable capacitor

Q41. A resistor has the color bands brown, black, orange, and gold. During measurement with a multimeter, the value appears close to 9.8 k Ω . Which conclusion is most appropriate? [Tag: Familiarization of electronic components | Hard]

(a) The gold band represents a temperature coefficient
(b) The resistor should measure near 100 Ω
(c) The resistor is within its stated tolerance
(d) The resistor must be connected with reverse polarity

Q42. A capacitor marked 104 J is connected to a circuit. Which interpretation is correct? [Tag: Familiarization of electronic components | Hard]

(a) 10.4 μ F with ± 5% tolerance
(b) 104 pF with ± 10% tolerance
(c) 100 nF with ± 5% tolerance
(d) 1 nF with ± 20% tolerance

Q43. A polarized electrolytic capacitor is installed across a DC supply with its negative terminal connected to the positive rail. What is the most likely consequence? [Tag: Familiarization of electronic components | Hard]

(a) A permanent increase in dielectric strength
(b) Increased leakage and possible capacitor failure
(c) Reduced capacitance with no other effect
(d) Improved ripple filtering at higher frequency

Q44. An LED and a 330 Ω resistor are connected in series to a 5 V supply. Assuming an LED forward drop of 2 V , which current is closest to the expected value? [Tag: Familiarization of electronic components | Hard]

(a) 9.1 mA
(b) 66.0 mA
(c) 6.1 mA
(d) 15.2 mA

Q45. A diode is tested using a digital multimeter in diode mode. It reads approximately 0.68 V in one direction and displays an open-circuit indication in the other. What does this most strongly indicate? [Tag: Familiarization of electronic components | Hard]

(a) A reverse-biased germanium diode
(b) A normally functioning silicon diode

(c) An open diode junction in both directions
(d) A shorted silicon diode junction

Q46. A transistor's data sheet identifies its terminals as E , B , and C , but the physical package orientation is unknown. Which method is safest for identifying the terminals before circuit connection? [Tag: Familiarization of electronic components | Hard]

(a) Use resistance mode between every pair only
(b) Connect the supply and identify terminals by heating
(c) Assume the center lead is always the base
(d) Use the data sheet and verify junctions with diode tests

Q47. A 5.1 V Zener diode is connected in reverse bias with a suitable series resistor. The supply is increased from 6 V to 12 V . Which behavior is expected if the current remains within its rated limits? [Tag: Familiarization of electronic components | Hard]

(a) The Zener voltage becomes approximately twice as large
(b) The Zener current remains exactly unchanged
(c) The Zener voltage remains approximately constant
(d) The Zener behaves as a forward-biased LED

Q48. A pushbutton input to a digital circuit changes unpredictably between logic states when pressed. Which component arrangement most directly addresses the problem? [Tag: Familiarization of electronic components | Hard]

(a) A fuse inserted in series with the input
(b) A larger LED connected across the input
(c) A reverse-biased diode placed across the supply
(d) A pull-up or pull-down resistor with switch debouncing

Q49. Two resistors are marked 1 k Ω and 2 k Ω . Which connection produces an equivalent resistance of approximately 667 Ω ? [Tag: Familiarization of electronic components | Hard]

(a) Connection with one resistor open
(b) Series connection
(c) Connection across the same wire
(d) Parallel connection

Q50. Before wiring a digital trainer circuit, the supply is set to the required voltage. Which additional practice best reduces the risk of damaging an integrated circuit? [Tag: Familiarization of the digital trainer module | Hard]

(a) Increase the supply voltage to improve noise immunity
(b) Verify polarity and current-limit settings before insertion
(c) Short unused IC inputs to the positive rail
(d) Insert the IC while the trainer output is enabled

Q51. A logic trainer's LED output indicator remains off when the measured output voltage is 3.8 V . Which diagnosis is most reasonable? [Tag: Familiarization of the digital trainer module | Hard]

(a) The output voltage proves the LED is open-circuit
(b) The IC must be operating without any supply voltage
(c) The logic output is definitely at binary zero
(d) The indicator may be active-low or incorrectly referenced

Q52. A digital trainer provides a clock generator and a manual pulse switch. A counter advances several times for one manual press. What is the most likely cause? [Tag: Familiarization of the digital trainer module | Hard]

(a) The display decoder has changed the counter state
(b) Mechanical switch bounce creates multiple clock edges
(c) The counter has converted binary data into analog data
(d) The supply voltage is necessarily below the logic threshold

Q53. A trainer module contains several unused inputs on a logic gate IC. Why should these inputs generally be assigned a defined logic level? [Tag: Familiarization of the digital trainer module | Hard]

(a) Unused inputs automatically increase the output drive current
(b) Defined inputs convert the gate into an analog amplifier
(c) Floating inputs can respond to noise and cause unstable operation
(d) Floating inputs always force the output permanently low

Q54. A trainer uses a common-ground digital supply. An oscilloscope probe is connected with its ground clip to a signal point rather than circuit ground. What is the primary risk? [Tag: Familiarization of the digital trainer module | Hard]

(a) The probe will convert the signal into a differential measurement
(b) The circuit will become immune to ground noise
(c) The probe ground may short the signal to earth or circuit ground
(d) The displayed waveform will automatically double in frequency

Q55. A logic trainer's regulated output is correct with no load but drops significantly when an IC circuit is connected. Which check should be performed first? [Tag: Familiarization of the digital trainer module | Hard]

(a) Disconnect the ground to remove loading
(b) Replace every resistor with a lower resistance
(c) Increase the regulated voltage above the IC rating
(d) Inspect for wiring errors or an unintended short circuit

Q56. A function generator is set to produce a 4 V pp sine wave with a 1 V DC offset. What are the maximum and minimum voltages relative to ground? [Tag: Familiarization of input sources and display devices related to electronic circuits | Hard]

(a) + 2 V and - 2 V
(b) + 5 V and - 3 V
(c) + 3 V and - 1 V
(d) + 4 V and 0 V

Q57. A function generator has a 50 Ω output and is connected to a high-impedance oscilloscope input. The generator display is calibrated for a 50 Ω load. What reading will commonly appear on the oscilloscope? [Tag: Familiarization of input sources and display devices related to electronic circuits | Hard]

(a) Approximately twice the displayed amplitude
(b) Approximately the same amplitude with reversed polarity
(c) Approximately half the displayed amplitude
(d) Approximately one-quarter the displayed amplitude

Q58. An oscilloscope displays a stable waveform only when its trigger level is set near the waveform's midpoint. What does this setting primarily control? [Tag: Familiarization of input sources and display devices related to electronic circuits | Hard]

(a) The input impedance selected by the probe
(b) The vertical scale used to measure waveform amplitude
(c) The frequency generated by the signal source
(d) The voltage threshold at which sweep acquisition begins

Q59. A seven-segment display is connected to a decoder whose outputs are active-low. The display segments illuminate only when their cathodes are driven low. Which display type is compatible? [Tag: Familiarization of input sources and display devices related to electronic circuits | Hard]

(a) Differential-output display
(b) Common-cathode display
(c) Common-anode display
(d) Open-collector sensor display

Q60. A digital circuit drives an LED directly from a logic output. The LED is brighter than expected and the IC becomes warm. Which change is most appropriate? [Tag: Familiarization of input sources and display devices related to electronic circuits | Hard]

(a) Remove the LED's polarity check from the circuit

(b) Connect the LED directly across the supply rails
(c) Add a correctly sized series current-limiting resistor
(d) Increase the logic output voltage above its rating

1.2 Section B: Comprehensive Subjective Problem with Detailed Analytical Rubric

Problem 1 **[Workshop Practical / Viva Scenario — 5/10 Marks]**

1 A carbon-film resistor carries four color bands in order: Yellow, Violet, Orange, and Gold. Determine its nominal resistance, upper and lower tolerance limits, and explain why measuring this resistor while still connected inside a powered-on circuit will lead to erroneous DMM readings.

Step-by-Step Analytical / Diagnostic Solution:

1. Color Code Decoding:
Yellow corresponds to digit 4, Violet to digit 7, and Orange corresponds to multiplier $10^3 = 1000\,\Omega$. The fourth band (Gold) represents a tolerance of $\pm 5\%$.
Nominal Resistance: $R = 47 \times 10^3\,\Omega = 47\,\text{k}\Omega$.
Tolerance range: $\pm 5\% \times 47\,\text{k}\Omega = \pm 2.35\,\text{k}\Omega$.
Upper Limit: $R_{\text{max}} = 47 + 2.35 = 49.35\,\text{k}\Omega$.
Lower Limit: $R_{\text{min}} = 47 - 2.35 = 44.65\,\text{k}\Omega$.

2. In-Circuit Measurement Error Mechanism:
When a DMM is set to resistance mode, it injects an internal calibrated test current I_{test} and measures the resultant terminal voltage $V = I_{\text{test}} R$. If the circuit is powered on or if parallel circuit paths exist:
(a) External circuit potentials corrupt the DMM's high-impedance voltage measurement ADC, potentially destroying the meter input fuse.
(b) Any parallel branches shunt a fraction of the test current away from the target resistor, yielding an equivalent parallel resistance $R_{\text{meas}} = R \parallel R_{\text{parallel}} < R$. Hence, resistors must always be tested de-energized and with at least one leg lifted from the breadboard.

Examiner Rubric (10 Marks):
- Correct decoding of color bands to digits & multiplier: 3 Marks
- Correct nominal value and $\pm 5\%$ tolerance band calculation: 2 Marks
- Detailed physical explanation of DMM constant-current ohmmeter circuit: 3 Marks
- Discussion of parallel loading paths and breadboard isolation rule: 2 Marks

1.3 Section C: Laboratory Viva-Voce Questions with Authoritative Solutions

Viva Question: What is the primary difference between passive and active electronic components?
Technical Answer: Passive components (resistors, capacitors, inductors) cannot amplify electrical signal power and do not require external power supplies to function. Active components (diodes, bipolar transistors, operational amplifiers) can control electron flow and amplify power, requiring an external DC operating bias.

Viva Question: Why does a Digital Storage Oscilloscope (DSO) require high input impedance (typically 1 MΩ)?
Technical Answer: A 1 MΩ input impedance minimizes the loading effect on the circuit under test, preventing the oscilloscope probe from drawing significant signal current and perturbing the observed waveform amplitude or timing.

Viva Question: What is the purpose of the current-limiting resistor connected in series with a Light Emitting Diode (LED)?
Technical Answer: An LED exhibits an exponential forward V-I characteristic. Once forward voltage exceeds cut-in ($V_F \approx 1.8$–2.2 V), internal dynamic resistance drops to near zero. Without a series resistor $R_S = \frac{V_{CC} - V_F}{I_F}$, current escalates uncontrolled, leading to immediate thermal destruction.

1.4 Section D: Workshop Troubleshooting & Fault Diagnostic Case Study

Troubleshooting Scenario: Unresponsive Logic Probe on Trainer Module
A student assembles a TTL logic circuit on the digital trainer module. However, the logic indicator LED remains permanently dark regardless of whether an input is tied to +5V or 0V.

Diagnostic Step 1: Check trainer DC power rails using DMM in DC voltage mode. Confirm $+5.0\ \text{V} \pm 0.25\ \text{V}$ between red rail and blue ground rail.
Diagnostic Step 2: Verify common ground connection. A classic workshop error is leaving the circuit ground unbonded to the trainer module ground rail, causing floating reference voltage.
Diagnostic Step 3: Check the LED polarity. The long lead (anode) must connect towards the positive logic output, with cathode to ground through the internal series limiting resistor.

Chapter 2

Unit 2: Voltage and Current Divider Laws & Prototyping Measurements

2.1 Section A: Multiple Choice Practice Questions (60 Curated Problems)

The following 60 multiple-choice questions reflect the official workshop evaluation bank, categorized by fundamental learning outcome and laboratory testing difficulty.

Q1. What is the main purpose of a voltage divider circuit? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) To store electrical charge
(b) To obtain a fraction of an input voltage
(c) To convert DC into AC
(d) To increase the input current

Q2. Which components are commonly used to form a basic voltage divider? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) Two resistors in series
(b) Two diodes in series
(c) Two capacitors in parallel
(d) Two inductors in parallel

Q3. For two series resistors, which formula gives the output voltage across R_2? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) $V_{\text{out}} = \frac{V_{\text{in}}}{R_1+R_2}$
(b) $V_{\text{out}} = V_{\text{in}}(R_1 + R_2)$
(c) $V_{\text{out}} = V_{\text{in}}\frac{R_1}{R_1+R_2}$
(d) $V_{\text{out}} = V_{\text{in}}\frac{R_2}{R_1+R_2}$

Q4. If $R_1 = R_2$ in a voltage divider, what is the output voltage across either resistor? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) V_{in}
(b) $\frac{V_{\text{in}}}{2}$
(c) 0
(d) $2V_{\text{in}}$

Q5. A voltage divider has V in = 12 V , R 1 = 2 k Ω , and R 2 = 4 k Ω . What is V o u t across R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) 2 V
(b) 8 V
(c) 12 V
(d) 4 V

Q6. In a series voltage divider, which resistor has the larger voltage drop? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) The resistor with the larger resistance
(b) Both always have equal drops
(c) The resistor connected to ground
(d) The resistor with the smaller resistance

Q7. What happens to the output voltage across R 2 if R 2 increases while R 1 and V in remain constant? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) It becomes zero
(b) It increases
(c) It decreases
(d) It remains unchanged

Q8. What is the total resistance of two series resistors R_1 and R_2? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) $R_1 R_2$
(b) $R_1 - R_2$
(c) $\frac{R_1}{R_2}$
(d) $R_1 + R_2$

Q9. How can the voltage divider law be experimentally validated? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) Measure only the resistor colors
(b) Disconnect the voltage source first
(c) Compare calculated and measured output voltages
(d) Replace both resistors with wires

Q10. Where should a voltmeter be connected to measure the divider output across R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Easy]

(a) Across R 2
(b) Across the voltage source only
(c) In series with R 2
(d) In series with the source

Q11. What is the main purpose of a current divider circuit? [Tag: Validation and analysis of the Current Divider Law | Easy]

(a) To split current among parallel branches
(b) To combine separate voltage sources
(c) To store current in a capacitor
(d) To increase total resistance only

Q12. Which connection is required for a basic current divider? [Tag: Validation and analysis of the Current Divider Law | Easy]

(a) Resistors connected in parallel
(b) Resistors connected without a source
(c) Resistors connected to open circuits
(d) Resistors connected in series

Q13. For two parallel resistors, which formula gives the current through R_1? [Tag: Validation and analysis of the Current Divider Law | Easy]

(a) $I_1 = I_T \frac{R_2}{R_1+R_2}$
(b) $I_1 = \frac{I_T}{R_1+R_2}$
(c) $I_1 = I_T(R_1 + R_2)$
(d) $I_1 = I_T \frac{R_1}{R_1+R_2}$

Q14. In a parallel circuit, which branch carries more current? [Tag: Validation and analysis of the Current Divider Law | Easy]

(a) Both branches always carry zero current
(b) The branch with larger resistance
(c) The branch with no voltage
(d) The branch with smaller resistance

Q15. If two parallel resistors have equal resistance, how is the total current divided? [Tag: Validation and analysis of the Current Divider Law | Easy]
(a) All current flows through one branch
(b) The current doubles in both branches
(c) The current divides equally
(d) No current enters either branch

Q16. A total current of 6 A enters two equal parallel resistors. What current flows through each resistor? [Tag: Validation and analysis of the Current Divider Law | Easy]
(a) 6 A
(b) 2 A
(c) 1 A
(d) 3 A

Q17. A current divider has $I_T = 6$ A, $R_1 = 2\ \Omega$, and $R_2 = 4\ \Omega$. What is the current through R_1? [Tag: Validation and analysis of the Current Divider Law | Easy]
(a) 2 A
(b) 6 A
(c) 1 A
(d) 4 A

Q18. What is the relationship between total current and branch currents in a two-branch parallel circuit? [Tag: Validation and analysis of the Current Divider Law | Easy]
(a) $I_T = I_1 + I_2$
(b) $I_T = I_1 - I_2$
(c) $I_T = \frac{I_1}{I_2}$
(d) $I_T = I_1 I_2$

Q19. How can the current divider law be experimentally validated? [Tag: Validation and analysis of the Current Divider Law | Easy]
(a) Measure only the supply voltage
(b) Connect the branches in series
(c) Compare calculated and measured branch currents
(d) Remove the parallel resistors

Q20. Where should an ammeter be connected to measure the current through one branch? [Tag: Validation and analysis of the Current Divider Law | Easy]
(a) Across the voltage source
(b) Across the entire circuit
(c) In series with the branch
(d) In parallel with the branch

Q21. A 12 V source is connected across two series resistors, $R_1 = 2$ kΩ and $R_2 = 4$ kΩ. What is the voltage across R_2? [Tag: Validation and analysis of the Voltage Divider Law | Medium]
(a) 6 V
(b) 4 V
(c) 10 V
(d) 8 V

Q22. A divider uses $R_1 = 3$ kΩ and $R_2 = 2$ kΩ across a 10 V supply. If the measured output across R_2 is 3.8 V, what is the percentage error relative to the theoretical value? [Tag: Validation and analysis of the Voltage Divider Law | Medium]
(a) 7.5%

(b) 2.5%
(c) 5.0%
(d) 10.0%

Q23. A 12 V divider has R 1 = 2 k Ω and R 2 = 6 k Ω . A 6 k Ω load is connected in parallel with R 2 . What is the loaded output voltage? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) 8.0 V
(b) 6.0 V
(c) 7.2 V
(d) 9.0 V

Q24. A voltage divider must produce 5 V from a 15 V source, with the output taken across R 2 . Which resistance relationship is required? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) R 1 = 3 R 2
(b) R 2 = 2 R 1
(c) R 1 = R 2
(d) R 1 = 2 R 2

Q25. Two series resistors, 1 k Ω and 2 k Ω , are connected to a 9 V source. Which pair of measured voltage drops would validate the voltage divider law? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) 6 V and 3 V
(b) 2 V and 7 V
(c) 4 V and 5 V
(d) 3 V and 6 V

Q26. Which voltmeter characteristic most effectively reduces loading error when experimentally validating a voltage divider? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) Low input resistance
(b) Low voltage range
(c) High current range
(d) High input resistance

Q27. A 10 V divider uses two nominal 10 k Ω resistors with ± 1% tolerance. What is the approximate minimum output across the lower resistor due only to resistor tolerance? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) 4.90 V
(b) 5.00 V
(c) 4.95 V
(d) 5.05 V

Q28. An 18 V divider has R 1 = 3 k Ω and R 2 = 6 k Ω . A 4 k Ω load is connected to its output. Using the divider's Thevenin equivalent, what is the loaded output voltage? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) 12 V
(b) 6 V
(c) 10 V
(d) 8 V

Q29. A 20 V source supplies series resistors R 1 = 2 k Ω and R 2 = 3 k Ω . What power is dissipated by R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) 32 mW
(b) 60 mW
(c) 40 mW
(d) 48 mW

Q30. The theoretical output of a divider is 6.00 V , while the measured output is 5.82 V . What is the magnitude of the percentage deviation? [Tag: Validation and analysis of the Voltage Divider Law | Medium]

(a) 3%
(b) 4%
(c) 5%
(d) 2%

Q31. A total current of 6 A enters two parallel resistors of 6 Ω and 3 Ω . What current flows through the 6 Ω resistor? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 4 A
(b) 1 A
(c) 2 A
(d) 3 A

Q32. Two parallel branches have conductances of 2 mS and 6 mS . If the total current is 16 mA , what current flows through the 6 mS branch? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 12 mA
(b) 8 mA
(c) 4 mA
(d) 14 mA

Q33. A total current of 6 mA enters three parallel resistors of 2 k Ω , 3 k Ω , and 6 k Ω . What current flows through the 3 k Ω branch? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 2 mA
(b) 3 mA
(c) 4 mA
(d) 1 mA

Q34. A current of 10 mA enters a parallel combination of 1 k Ω and 4 k Ω . What is the voltage across the parallel network? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 10 V
(b) 4 V
(c) 6 V
(d) 8 V

Q35. During validation of a parallel circuit, the total current is measured as 8.0 mA , while the branch currents are 5.1 mA and 2.8 mA . What is the percentage discrepancy relative to the total current? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 1.25%
(b) 0.80%
(c) 2.50%
(d) 3.75%

Q36. A total current of 12 mA divides between parallel resistors of 2 k Ω and 4 k Ω . What power is dissipated by the 4 k Ω resistor? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 96 mW
(b) 48 mW
(c) 64 mW
(d) 32 mW

Q37. A total current of 5 mA enters two parallel branches. The first branch contains 2 k Ω and carries 3 mA . What is the resistance of the second branch? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 1 k Ω
(b) 2 k Ω

(c) 4 k Ω
(d) 3 k Ω

Q38. Under which circuit condition can the current divider law be directly applied? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) Resistors dissipate the same power
(b) Resistors carry the same series current
(c) Resistors share the same two nodes
(d) Resistors have equal voltage ratings

Q39. An ideal voltage source supplies parallel branches of 2 k Ω and 3 k Ω . An ammeter with 100 Ω internal resistance is inserted into the 2 k Ω branch. What happens to the measured circuit currents? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) Both branch currents become equal
(b) The second and total currents increase
(c) Both branch currents remain unchanged
(d) The first and total currents decrease

Q40. A 12 mA current divides between a 4 k Ω resistor and an unknown parallel resistor. If the 4 k Ω branch carries 3 mA , what is the unknown resistance? [Tag: Validation and analysis of the Current Divider Law | Medium]

(a) 2.00 k Ω
(b) 3.00 k Ω
(c) 1.00 k Ω
(d) 1.33 k Ω

Q41. An 18 V source feeds R 1 = 2 k Ω in series with R 2 = 4 k Ω . A 12 k Ω load is connected across R 2 . What is the loaded output voltage across the load? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 6.00 V
(b) 7.20 V
(c) 8.00 V
(d) 9.00 V

Q42. A 10 V source is connected to R 1 = 100 k Ω and R 2 = 900 k Ω . A voltmeter with input resistance 1 M Ω measures the voltage across R 2 . What reading should be expected? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 7.50 V
(b) 8.75 V
(c) 8.26 V
(d) 9.00 V

Q43. For a divider with fixed input voltage and fixed total resistance R 1 + R 2 = 10 k Ω , which choice maximizes the power dissipated in R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) R 1 = 5 k Ω , R 2 = 5 k Ω
(b) R 1 = 4 k Ω , R 2 = 6 k Ω
(c) R 1 = 2 k Ω , R 2 = 8 k Ω
(d) R 1 = 8 k Ω , R 2 = 2 k Ω

Q44. A divider has R 1 = 1 k Ω and R 2 = 2 k Ω , each with a tolerance of ± 5% . What are the worst-case limits of V o / V i when the output is taken across R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 0.633 to 0.667
(b) 0.615 to 0.680
(c) 0.600 to 0.700
(d) 0.603 to 0.689

Q45. A 12 V source feeds R 1 = 3 k Ω and R 2 = 3 k Ω . A 6 k Ω load is connected across R 2 . What is the output voltage? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 4.00 V
(b) 3.60 V
(c) 6.00 V
(d) 4.80 V

Q46. A divider is driven by an ideal 5 V source through a source resistance of 50 Ω , followed by R 1 = 950 Ω and R 2 = 1 k Ω . What voltage appears across R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 2.63 V
(b) 2.44 V
(c) 2.56 V
(d) 2.50 V

Q47. A divider uses R 1 = 2 k Ω , R 2 = 3 k Ω , and a load R L . With V i = 15 V , the desired loaded output is 6 V . What value of R L is required? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 2.40 k Ω
(b) 1.20 k Ω
(c) 1.80 k Ω
(d) 3.60 k Ω

Q48. A voltage divider has R 1 = 4 k Ω and R 2 = 6 k Ω . If a finite-input-resistance voltmeter changes the measured output from the ideal value by less than 1% , what is the minimum approximate meter resistance? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 60 k Ω
(b) 600 k Ω
(c) 300 k Ω
(d) 120 k Ω

Q49. For an unloaded divider, V o = V i R 2 / (R 1 + R 2) . If R 1 is increased by 10% while R 2 remains fixed, with R 1 = R 2 initially, what is the percentage change in V o ? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) A decrease of approximately 9.1%
(b) An increase of approximately 9.1%
(c) A decrease of approximately 4.8%
(d) An increase of approximately 4.8%

Q50. A practical divider is connected to a 20 V source with internal resistance 100 Ω . The divider consists of R 1 = 900 Ω and R 2 = 1 k Ω . What is the actual output across R 2 ? [Tag: Validation and analysis of the Voltage Divider Law | Hard]

(a) 9.52 V
(b) 9.00 V
(c) 10.00 V
(d) 10.53 V

Q51. A total current of 6 mA enters three parallel resistors of 2 k Ω , 3 k Ω , and 6 k Ω . What current flows through the 2 k Ω branch? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 1.0 mA
(b) 2.0 mA
(c) 4.0 mA
(d) 3.0 mA

Q52. A 10 mA current source feeds parallel branches of 4 k Ω and 6 k Ω . What are the currents in the 4 k Ω and 6 k Ω branches, respectively? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 7.5 mA and 2.5 mA

(b) 5 mA and 5 mA
(c) 6 mA and 4 mA
(d) 4 mA and 6 mA

Q53. A 3 mA ideal current source feeds three parallel resistances: 10 k Ω , 20 k Ω , and 30 k Ω . What current flows through the 10 k Ω branch? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 2.00 mA
(b) 1.64 mA
(c) 1.80 mA
(d) 1.09 mA

Q54. A 12 A current enters two parallel paths. One path is 4 Ω ; the other consists of 6 Ω in series with 2 Ω . What current flows through the 2 Ω resistor? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 8 A
(b) 4 A
(c) 6 A
(d) 3 A

Q55. Two parallel branches have resistances 8 Ω and 12 Ω . If the current in the 8 Ω branch is 3 A , what is the total source current? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 4.5 A
(b) 5.0 A
(c) 7.5 A
(d) 4.0 A

Q56. A current divider has two branches, R 1 = 5 k Ω and R 2 = 15 k Ω . If the total current is 8 mA , what is the current through R 2 ? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 6.0 mA
(b) 1.0 mA
(c) 4.0 mA
(d) 2.0 mA

Q57. A 2 A source supplies two parallel branches. Branch A has 3 Ω and branch B has 6 Ω . An ammeter with 0.5 Ω burden resistance is inserted only in branch A. What current does the ammeter indicate? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 0.80 A
(b) 1.00 A
(c) 1.33 A
(d) 1.14 A

Q58. Three equal resistors are connected in parallel to an ideal current source. One resistor is then short-circuited by an ideal wire. What happens to the current in each resistor? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) All three retain one-third of the source current
(b) The current divides equally between the two unshorted resistors
(c) The shorted resistor carries half the current; the others share half
(d) The shorted resistor carries all current; the others carry zero

Q59. A total current I_T enters two parallel resistors R_1 and R_2. Which expression gives the current through R_1 and remains valid when $R_1 \neq R_2$? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) $I_1 = I_T \frac{R_1 R_2}{R_1+R_2}$
(b) $I_1 = I_T \frac{R_1+R_2}{R_2}$
(c) $I_1 = I_T \frac{R_1}{R_1+R_2}$
(d) $I_1 = I_T \frac{R_2}{R_1+R_2}$

Q60. A 6 mA current source feeds two parallel branches: R 1 = 2 k Ω and R 2 = 4 k Ω . A 6 k Ω resistor is connected in parallel with R 2 . What current flows through the original R 1 branch? [Tag: Validation and analysis of the Current Divider Law | Hard]

(a) 3.00 mA
(b) 4.00 mA
(c) 3.60 mA
(d) 2.00 mA

2.2 Section B: Comprehensive Subjective Problem with Detailed Analytical Rubric

Problem 2 **[Workshop Practical / Viva Scenario — 5/10 Marks]**

1 A series circuit consists of $V_{\text{in}} = 12\,\text{V}$, $R_1 = 10\,\text{k}\Omega$, and $R_2 = 10\,\text{k}\Omega$. (a) Calculate the theoretical unloaded voltage across R_2. (b) If an analog voltmeter with internal resistance $R_m = 20\,\text{k}\Omega$ is connected across R_2, determine the measured voltage and the resulting percentage loading error. (c) What is the required meter impedance to keep loading error below 1%?

Step-by-Step Analytical / Diagnostic Solution:

(a) Unloaded Output Voltage:
By the Voltage Divider Law:

$$V_2 = V_{\text{in}} \frac{R_2}{R_1 + R_2} = 12\,\text{V} \times \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 10\,\text{k}\Omega} = 12 \times \frac{1}{2} = 6.00\,\text{V}$$

(b) Loaded Output Voltage:
Connecting the meter across R_2 places R_m in parallel with R_2:

$$R_{2,\text{eq}} = R_2 \parallel R_m = \frac{R_2 \cdot R_m}{R_2 + R_m} = \frac{10 \times 20}{10 + 20}\,\text{k}\Omega = \frac{200}{30} = 6.67\,\text{k}\Omega$$

The loaded divider voltage is:

$$V_{2,\text{meas}} = V_{\text{in}} \frac{R_{2,\text{eq}}}{R_1 + R_{2,\text{eq}}} = 12 \times \frac{6.67}{10 + 6.67} = 12 \times \frac{6.67}{16.67} = 4.80\,\text{V}$$

Loading Error:

$$\%\ \text{Error} = \frac{V_{\text{true}} - V_{\text{meas}}}{V_{\text{true}}} \times 100\% = \frac{6.00 - 4.80}{6.00} \times 100\% = 20.0\%$$

(c) High-Impedance Condition for $\leq 1\%$ Error:
For error $\leq 1\%$, $V_{\text{meas}} \geq 5.94\,\text{V}$. Setting $V_{\text{meas}} = 12\frac{R_{2,\text{eq}}}{10+R_{2,\text{eq}}} \geq 5.94$ yields $R_{2,\text{eq}} \geq 9.80\,\text{k}\Omega$, requiring $R_m \geq \frac{10\times 9.80}{10-9.80} \approx 490\,\text{k}\Omega$. Modern DMMs have $R_m = 10\,\text{M}\Omega$, ensuring negligible error ($< 0.05\%$).

Examiner Rubric (10 Marks):
- Unloaded voltage derivation & correct value: 2 Marks
- Equivalent parallel resistance $R_2 \parallel R_m$: 3 Marks
- Loaded voltage computation & percentage error: 3 Marks
- Analysis of meter impedance threshold for precision measurement: 2 Marks

2.3 Section C: Laboratory Viva-Voce Questions with Authoritative Solutions

Viva Question: State the dual relationship between the Voltage Divider Law and the Current Divider Law.
Technical Answer: The Voltage Divider Law applies to series circuits where current is constant and source voltage distributes proportionally to individual resistances: $V_x = V_T(R_x/R_{\text{total}})$. The Current Divider Law applies to parallel circuits where voltage is constant and total current distributes proportionally to individual conductances: $I_x = I_T(G_x/G_{\text{total}}) = I_T(R_p/R_x)$.

Viva Question: Why does a potentiometer act as a variable voltage divider?
Technical Answer: A potentiometer consists of a resistive track of fixed length with a sliding wiper contact. The wiper divides the track into two series resistors R_1 and R_2 whose sum is constant ($R_1 + R_2 = R_{\text{pot}}$). Moving

the wiper adjusts the ratio $R_2/(R_1 + R_2)$ continuously from 0 to 1, producing an output voltage from 0 V to V_{CC}.

Viva Question: What is ammeter burden voltage?
Technical Answer: Ammeter burden voltage is the small voltage drop developed across the internal current-sensing shunt resistor when current flows through the meter. In low-voltage circuits, this burden voltage reduces total branch current, introducing measurement error.

2.4 Section D: Workshop Troubleshooting & Fault Diagnostic Case Study

Troubleshooting Scenario: Voltage Divider Output Drops When Load Connected
A student constructs a 5V to 2.5V reference voltage divider using two $100\,\text{k}\Omega$ resistors from a 5V supply. When an external circuit with an input resistance of $20\,\text{k}\Omega$ is attached to the output, the voltage collapses to 0.83V.

Diagnosis: The student failed to account for source output impedance (Thevenin equivalent resistance $R_{th} = 100\,\text{k}\Omega \parallel 100\,\text{k}\Omega = 50\,\text{k}\Omega$). Connecting a low load impedance ($20\,\text{k}\Omega$) severely loads the divider.
Remedy: Either decrease divider resistors to $R_1 = R_2 = 200\,\Omega$ (at the cost of higher quiescent power consumption) or place an op-amp unity-gain voltage follower (buffer) between the divider and load.

Chapter 3

Unit 3: Semiconductor PN Junction Diode Forward Characteristics & Applications

3.1 Section A: Multiple Choice Practice Questions (60 Curated Problems)

The following 60 multiple-choice questions reflect the official workshop evaluation bank, categorized by fundamental learning outcome and laboratory testing difficulty.

Q1. What does forward biasing a PN junction diode mean? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) Connecting both sides to the positive terminal
(b) Connecting P-side to negative and N-side to positive
(c) Connecting both sides to the negative terminal
(d) Connecting P-side to positive and N-side to negative

Q2. What happens to the depletion-region width when a PN junction diode is forward biased? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) It becomes infinite
(b) It becomes narrower
(c) It becomes wider
(d) It remains unchanged

Q3. What happens to the barrier potential of a PN junction under forward bias? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) It remains constant
(b) It increases
(c) It decreases
(d) It reverses polarity

Q4. Which charge carriers mainly contribute to current in a forward-biased PN junction diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) Stationary ions
(b) Majority carriers
(c) Minority carriers
(d) Neutral atoms

Q5. What is the typical cut-in voltage of a silicon diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) 2.0 V
(b) 0.3 V

(c) 0.7 V
(d) 1.2 V

Q6. What is the typical cut-in voltage of a germanium diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) 1.5 V
(b) 1.0 V
(c) 0.7 V
(d) 0.3 V

Q7. What is another common name for the cut-in voltage of a diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) Reverse voltage
(b) Breakdown voltage
(c) Knee voltage
(d) Peak voltage

Q8. How does forward current change after the diode voltage exceeds the knee voltage? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) It rises rapidly
(b) It becomes zero
(c) It falls rapidly
(d) It stays constant

Q9. Which axis normally represents diode current on a voltage-current characteristic graph? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) Circular axis
(b) Diagonal axis
(c) Horizontal axis
(d) Vertical axis

Q10. Which axis normally represents diode voltage on a voltage-current characteristic graph? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) Horizontal axis
(b) Diagonal axis
(c) Vertical axis
(d) Circular axis

Q11. In which quadrant is the forward-bias characteristic of a diode usually plotted? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) Third quadrant
(b) Fourth quadrant
(c) First quadrant
(d) Second quadrant

Q12. How does an ideal diode behave when it is forward biased? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) As a closed switch
(b) As a current source
(c) As an open switch
(d) As a charged capacitor

Q13. What is the forward-voltage drop across an ideal conducting diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]
(a) 0.3 V
(b) 0 V
(c) 0.7 V
(d) 1.0 V

Q14. Before the forward voltage reaches the knee voltage, the diode current is generally: [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) Very large
(b) Very small
(c) Exactly infinite
(d) Constantly negative

Q15. Which instrument is connected in series to measure the forward current of a diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) Wattmeter
(b) Ammeter
(c) Voltmeter
(d) Ohmmeter

Q16. Which instrument is connected across a diode to measure its forward voltage? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) Ammeter
(b) Voltmeter
(c) Galvanometer
(d) Frequency meter

Q17. Why is a resistor commonly connected in series with a forward-biased diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) To create breakdown
(b) To limit current
(c) To increase voltage
(d) To reverse polarity

Q18. The static forward resistance of a diode at an operating point is calculated using which expression? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) $R_F = V_F \cdot I_F$
(b) $R_F = V_F + I_F$
(c) $R_F = \frac{I_F}{V_F}$
(d) $R_F = \frac{V_F}{I_F}$

Q19. A diode has a forward voltage of 0.6 V and a forward current of 3 mA . What is its static forward resistance? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) 200 Ω
(b) 300 Ω
(c) 600 Ω
(d) 100 Ω

Q20. What does the nonlinear shape of a diode's forward characteristic indicate? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Easy]

(a) Resistance remains perfectly constant
(b) Current is not directly proportional to voltage
(c) Current is always equal to voltage
(d) Voltage remains zero at all currents

Q21. A silicon PN junction diode is forward biased. At which voltage does its current typically begin to increase significantly? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) About 0.5 V
(b) About 0.7 V
(c) About 1.2 V
(d) About 0.3 V

Q22. For a silicon diode operating near its knee voltage, what is the most likely effect of increasing the forward voltage from 0.7 V to 0.8 V ? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The diode becomes reverse biased
(b) The current decreases slightly
(c) The current increases substantially
(d) The current remains exactly constant

Q23. The forward current of a diode changes from 2 mA to 3 mA when its forward voltage changes from 0.68 V to 0.70 V . What is its approximate dc or small-signal resistance over this interval? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) 200 Ω
(b) 50 Ω
(c) 5 Ω
(d) 20 Ω

Q24. A diode has a forward voltage of 0.72 V and carries a current of 15 mA . What power is dissipated in the diode? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) 10.8 mW
(b) 48 mW
(c) 1.08 mW
(d) 108 mW

Q25. When the temperature of a silicon diode increases, how does its forward characteristic generally change at a fixed forward current? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The characteristic becomes vertical
(b) The required voltage increases
(c) The required voltage decreases
(d) The current becomes zero

Q26. A 5 V source is connected in series with a 1 k Ω resistor and a silicon diode modeled with a constant forward drop of 0.7 V . What is the approximate diode current? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) 0.43 mA
(b) 4.3 mA
(c) 7.0 mA
(d) 5.7 mA

Q27. On a forward-bias characteristic curve, the current changes from 4 mA to 6 mA while the voltage changes from 0.64 V to 0.66 V . What is the approximate slope Δ I /Δ V ? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) 10 S
(b) 100 S
(c) 0.1 S
(d) 50 S

Q28. Which feature best identifies the knee region of a PN junction diode's forward characteristic? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) Current changes rapidly after a small voltage rise
(b) Voltage and current both remain constant
(c) Voltage changes rapidly after a small current rise
(d) Current remains zero for every voltage

Q29. Compared with a silicon diode, a germanium diode generally shows which forward-bias

characteristic? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) A higher cut-in voltage
(b) An identical cut-in voltage
(c) A lower cut-in voltage
(d) No forward current

Q30. Using the diode equation with the same saturation current and ideality factor, the forward voltage increases from 0.60 V to 0.65 V at room temperature. Approximately how does the current change for n = 1 ? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) It becomes about half
(b) It remains nearly unchanged
(c) It increases by exactly 25 times
(d) It increases by about 7.4 times

Q31. For a forward-biased diode, which region of the characteristic is most accurately modeled by an approximately constant voltage drop? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The region after the knee voltage
(b) The region near reverse breakdown
(c) The region before any applied voltage
(d) The region below zero current

Q32. If the forward voltage of a diode is plotted on the horizontal axis and forward current on the vertical axis, what does a steep curve indicate? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) A small dynamic resistance
(b) A zero saturation current
(c) A reverse-biased condition
(d) A large dynamic resistance

Q33. For a diode with ideality factor n = 2 at room temperature, approximately what forward-voltage increase is required to increase the current by a factor of ten? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) 120 mV
(b) 240 mV
(c) 60 mV
(d) 30 mV

Q34. A measured diode characteristic shows a nearly straight line with a nonzero slope after the knee region. What does the slope mainly represent? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The diode's reverse voltage
(b) The diode's series resistance
(c) The diode's junction capacitance
(d) The diode's breakdown current

Q35. At room temperature, a diode carries 10 mA and has ideality factor n = 1 . Using r d ≈ n V T / I D , what is its approximate dynamic resistance? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) 25 Ω
(b) 2.5 Ω
(c) 0.25 Ω
(d) 250 Ω

Q36. A forward-biased diode has a current of 1 mA at 0.60 V and 10 mA at 0.66 V . Which conclusion is most reasonable at room temperature? [Tag: Analysis of the voltage-current characteristics

of a PN junction diode under forward-bias conditions | Medium]

(a) The ideality factor is close to 3
(b) The ideality factor is close to 1
(c) The diode has zero saturation current
(d) The diode is reverse biased

Q37. A diode and a resistor are connected to a dc source. If the source voltage increases while the diode remains in forward conduction, what primarily limits the increase in diode current? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The depletion layer alone
(b) The reverse saturation current
(c) The absence of forward bias
(d) The resistor in series

Q38. For a forward-biased silicon diode, the current is 5 mA at 2 5 ° C . If the temperature rises while the current is held constant, what happens to the forward voltage? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) It reverses polarity immediately
(b) It remains exactly unchanged
(c) It increases by about 2 mV / ° C
(d) It decreases by about 2 mV / ° C

Q39. A diode's forward characteristic is initially almost horizontal and then becomes steep. What physical interpretation best matches the steep portion? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The diode current is independent of voltage
(b) A small voltage change produces a large current change
(c) The diode is operating in reverse breakdown
(d) The junction barrier is increasing strongly

Q40. A load line intersects a diode's forward I - V characteristic at 0.68 V and 8 mA . What does this intersection represent? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Medium]

(a) The diode's reverse breakdown point
(b) The diode's zero-current voltage
(c) The circuit's forward operating point
(d) The maximum possible source voltage

Q41. At 300 K , a diode carries 0.10 mA at 0.50 V and 1.00 mA at 0.62 V . Assuming the Shockley model and negligible series resistance, what is the ideality factor n ? Use V T = 25.85 mV . [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) n ≈ 2.02
(b) n ≈ 1.01
(c) n ≈ 1.52
(d) n ≈ 2.51

Q42. A diode at 300 K carries 1.00 mA at 0.600 V . If n = 1 and the - 1 term is negligible, what is its reverse saturation current? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 2.41 × 1 0 - 13 A
(b) 1.20 × 1 0 - 12 A
(c) 8.30 × 1 0 - 14 A
(d) 4.15 × 1 0 - 14 A

Q43. A forward-biased diode operates at I D = 2.00 mA with n = 2 at 300 K . Neglecting series resistance and assuming I D ≫ I S , what is its small-signal resistance? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 51.70 Ω

(b) 25.85 Ω
(c) 103.4 Ω
(d) 12.93 Ω

Q44. A diode with n = 1 and V T = 25.0 mV is biased at 1.00 mA . Its voltage is changed by + 25.0 mV and then by - 25.0 mV relative to the bias point. What is the ratio of the upward current change to the magnitude of the downward current change? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 2.718
(b) 7.389
(c) 1.718
(d) 1.000

Q45. A diode with I S = 1 0 - 12 A and n = 1 is connected in series with 1.00 k Ω across a 5.00 V supply. At 300 K , which operating point most closely satisfies both the diode equation and the load line? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) I D ≈ 4.70 mA , V D ≈ 0.300 V
(b) I D ≈ 4.00 mA , V D ≈ 1.00 V
(c) I D ≈ 4.43 mA , V D ≈ 0.574 V
(d) I D ≈ 5.00 mA , V D ≈ 0.535 V

Q46. At 300 K , a diode with n = 1 has measured terminal voltages of 0.750 V at 10.0 mA and 1.000 V at 100 mA . Assuming a constant series resistance, estimate that resistance. [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 2.78 Ω
(b) 2.12 Ω
(c) 1.06 Ω
(d) 4.24 Ω

Q47. A diode carries a fixed current at 300 K with a forward voltage of 0.650 V . Its saturation current doubles for every 8 K rise, while n = 1 remains constant. According to the Shockley model, what is the forward voltage at 340 K ? Use V T = 25.85 mV at 300 K . [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 0.560 V
(b) 0.737 V
(c) 0.600 V
(d) 0.635 V

Q48. The straight-line region of a diode's forward characteristic plotted as V D versus lo g 10 I D has a slope of 89.3 mV per decade at 300 K . What ideality factor does this imply? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) n ≈ 1.00
(b) n ≈ 1.25
(c) n ≈ 1.50
(d) n ≈ 2.00

Q49. A practical diode has n = 2 , series resistance R s = 3.00 Ω , and a bias current of 10.0 mA at 300 K . What is its terminal small-signal conductance? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 0.0785 S
(b) 0.122 S
(c) 0.193 S
(d) 0.333 S

Q50. A diode with n = 1 is biased at 2.00 mA through a 100 Ω resistor. At 300 K , what is the small-signal change in diode current per volt of supply variation, assuming no diode series resistance? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 8.86 mA/V
(b) 77.4 mA/V
(c) 10.0 mA/V
(d) 7.97 mA/V

Q51. For a diode with I S = 10.0 nA , n = 2 , and V T = 25.85 mV , what current flows at a small forward voltage of 10.0 mV when the full Shockley equation is used? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 2.13 nA
(b) 10.0 nA
(c) 8.24 nA
(d) 12.1 nA

Q52. A diode operates at V D = 0.700 V and I D = 10.0 mA with n = 2 at 300 K . Neglecting series resistance, which pair gives its DC resistance and small-signal resistance, respectively? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 35.0 Ω and 5.17 Ω
(b) 5.17 Ω and 70.0 Ω
(c) 70.0 Ω and 5.17 Ω
(d) 70.0 Ω and 2.59 Ω

Q53. At 300 K , the forward current of a PN diode doubles whenever its voltage increases by approximately 36.0 mV over a certain low-current range. Which interpretation is most consistent with this observation? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) n ≈ 2 , suggesting recombination-dominated current
(b) n ≈ 1 , suggesting diffusion-dominated current
(c) n ≈ 4 , suggesting series-resistance domination
(d) n ≈ 0.5 , suggesting tunneling-dominated current

Q54. For a diode described by V D = n V T ln (I D / I S) + I D R s , the measured values of d V D / d ln I D are 45.85 mV at 10.0 mA and 125.85 mV at 50.0 mA . At 300 K , determine n and R s . [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) n = 1.0 and R s = 2.0 Ω
(b) n = 2.0 and R s = 2.0 Ω
(c) n = 2.0 and R s = 1.0 Ω
(d) n = 1.0 and R s = 4.0 Ω

Q55. Two parallel diodes have equal ideality factors and temperatures but saturation currents satisfying I S 1 = 4 I S 2 . If their total forward current is 10.0 mA and series resistances are negligible, how is the current shared? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) I 1 = 8.0 mA , I 2 = 2.0 mA
(b) I 1 = 5.0 mA , I 2 = 5.0 mA
(c) I 1 = 9.0 mA , I 2 = 1.0 mA
(d) I 1 = 6.0 mA , I 2 = 4.0 mA

Q56. Two parallel diodes each initially carry 1.00 mA at the same voltage. Their ideality factors are n 1 = 1 and n 2 = 2 . At 300 K , the common voltage is increased by 51.7 mV . Neglecting series resistance, what is the new total current? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 7.39 mA
(b) 14.78 mA
(c) 5.44 mA
(d) 10.11 mA

Q57. A diode's terminal voltage rises by 160 mV as its current increases from 1.00 mA to 10.0

mA . If its series resistance is 10.0 Ω , what ideality factor is implied at 300 K ? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) n ≈ 1.68
(b) n ≈ 2.69
(c) n ≈ 0.84
(d) n ≈ 1.18

Q58. A diode has n = 1.5 and an uncertain saturation current as high as 5.0 × 1 0 - 12 A . At 300 K , what is the maximum applied forward voltage that guarantees the Shockley current does not exceed 20.0 mA ? Neglect series resistance. [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 0.786 V
(b) 0.824 V
(c) 0.857 V
(d) 0.914 V

Q59. An ideal diode with n = 1 and V T = 25.0 mV is biased at 1.00 mA . A zero-average sinusoidal voltage v (t) = 10.0 sin (ω t) mV is superimposed. Neglecting I S relative to forward current, what is the resulting average current? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) 1.173 mA
(b) 1.040 mA
(c) 1.000 mA
(d) 1.020 mA

Q60. A diode carries 1.00 mA at both 300 K and 350 K . Its ideality factor is 1 , while I S changes from 1 0 - 12 A at 300 K to 1 0 - 9 A at 350 K . Using V T = 25.85 mV and 30.16 mV , respectively, what is the forward-voltage change? [Tag: Analysis of the voltage-current characteristics of a PN junction diode under forward-bias conditions | Hard]

(a) It decreases by approximately 119 mV
(b) It decreases by approximately 60 mV
(c) It increases by approximately 60 mV
(d) It increases by approximately 119 mV

3.2 Section B: Comprehensive Subjective Problem with Detailed Analytical Rubric

Problem 3 **[Workshop Practical / Viva Scenario — 5/10 Marks]**

1 A Silicon PN junction diode at room temperature ($T = 300\,\text{K}$, $V_T = 26\,\text{mV}$) has a reverse saturation current $I_s = 2\,\text{nA}$ and ideality factor $\eta = 1$. (a) Determine the diode forward current when $V_D = 0.65\,\text{V}$. (b) Calculate the static (DC) resistance and dynamic (AC) resistance at this operating point. (c) Explain why dynamic resistance is vastly smaller than static resistance in the forward conduction region.

Step-by-Step Analytical / Diagnostic Solution:

(a) Forward Conduction Current:
Using the Shockley diode equation:

$$I_D = I_s \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

Since $V_D/\eta V_T = 0.65/0.026 = 25 \gg 1$, the -1 term is negligible:

$$I_D \approx 2 \times 10^{-9} \times e^{25} = 2 \times 10^{-9} \times 7.20 \times 10^{10} = 144.0\,\text{mA}$$

(b) Static and Dynamic Resistances:
1. Static (DC) Resistance:

$$R_D = \frac{V_D}{I_D} = \frac{0.65\,\text{V}}{0.144\,\text{A}} \approx 4.51\,\Omega$$

2. Dynamic (AC) Resistance:

$$r_d = \frac{dV_D}{dI_D} = \frac{\eta V_T}{I_D} = \frac{0.026\,\text{V}}{0.144\,\text{A}} \approx 0.18\,\Omega$$

(c) Physical Interpretation:
The static resistance R_D is the reciprocal of the slope of the secant line from the origin $(0,0)$ to the Q-point (V_D, I_D). The dynamic resistance r_d is the reciprocal of the tangent slope at the Q-point. Due to the exponential nature of forward conduction, the slope $dI_D/dV_D = I_D/(\eta V_T)$ increases steeply with current, causing the incremental resistance r_d to become much smaller than R_D.

Examiner Rubric (10 Marks):
- Application of Shockley diode equation with correct exponential evaluation: 4 Marks
- Calculation of static resistance R_D: 2 Marks
- Derivation and numerical calculation of dynamic resistance $r_d = \eta V_T / I_D$: 2 Marks
- Graphical distinction between tangent and secant slopes: 2 Marks

3.3 Section C: Laboratory Viva-Voce Questions with Authoritative Solutions

Viva Question: What is the knee or cut-in voltage of a diode, and what causes it physically?
Technical Answer: The cut-in voltage ($V_k \approx 0.7$ V for Si, 0.3 V for Ge) is the forward bias potential required to counteract the built-in electrostatic potential barrier of the depletion region, allowing majority carriers to diffuse rapidly across the junction.

Viva Question: Why is a series current-limiting resistor mandatory during forward-bias diode testing?
Technical Answer: In the forward bias conduction region beyond cut-in, the diode dynamic resistance drops below $1\,\Omega$. Connecting a DC power supply directly across the diode without a series resistor leads to excessive

current exceeding maximum forward ratings ($I_{F,\max}$), causing immediate burnout.

Viva Question: How do you test a diode using a Digital Multimeter's Diode Check function?
Technical Answer: In forward bias (red lead to anode, black to cathode), the DMM injects ≈ 1 mA and displays the forward drop (0.55–0.75 V for Silicon). In reverse bias (red lead to cathode, black to anode), the display shows 'OL' (overload/infinite resistance). If it reads zero in both directions, the diode is shorted; if 'OL' in both, it is open.

3.4 Section D: Workshop Troubleshooting & Fault Diagnostic Case Study

Troubleshooting Scenario: Diode Circuit Emits Smoke in Rectifier Lab
During a bridge rectifier experiment, an engineer switches on the transformer secondary (15 $text V_{mathrm rms}$). Immediately, one 1N4007 diode overheats and cracks.

Diagnostic Cause: Diode polarity reversal. One diode was inserted backwards in the bridge configuration. When that half of the AC cycle occurred, the reversed diode and its counterpart created a direct dead short-circuit across the transformer secondary winding with zero series load resistance, drawing tens of amperes.
Preventative Workshop Rule: Always double check the cathode band (silver or white stripe) against the circuit schematic prior to energizing any power supply.

Chapter 4

Unit 4: Digital Logic Gates & Universal Gate Circuit Implementations

4.1 Section A: Multiple Choice Practice Questions (60 Curated Problems)

The following 60 multiple-choice questions reflect the official workshop evaluation bank, categorized by fundamental learning outcome and laboratory testing difficulty.

Q1. What is the main purpose of using a truth table while validating a logic gate? [Tag: Implementation and validation of logic gates | Easy]

(a) To compare expected and observed outputs
(b) To calculate the power supply frequency
(c) To measure the physical size of the gate
(d) To identify the color of connecting wires

Q2. How many possible input combinations must be tested for a two-input logic gate? [Tag: Implementation and validation of logic gates | Easy]

(a) Two combinations
(b) Eight combinations
(c) Four combinations
(d) Three combinations

Q3. For a two-input AND gate, which input combination produces a HIGH output? [Tag: Implementation and validation of logic gates | Easy]

(a) A = 1 , B = 1
(b) A = 1 , B = 0
(c) A = 0 , B = 0
(d) A = 0 , B = 1

Q4. For a two-input OR gate, which input combination produces a LOW output? [Tag: Implementation and validation of logic gates | Easy]

(a) A = 1 , B = 0
(b) A = 0 , B = 0
(c) A = 0 , B = 1
(d) A = 1 , B = 1

Q5. If the input of a NOT gate is HIGH, what should its output be during validation? [Tag: Implementation and validation of logic gates | Easy]

(a) LOW
(b) Unchanged
(c) Floating
(d) HIGH

Q6. When does a two-input NAND gate produce a LOW output? [Tag: Implementation and validation of logic gates | Easy]

(a) When both inputs are LOW
(b) When the inputs are different
(c) When either input is LOW
(d) When both inputs are HIGH

Q7. When does a two-input NOR gate produce a HIGH output? [Tag: Implementation and validation of logic gates | Easy]

(a) When both inputs are LOW
(b) When the inputs are different
(c) When either input is HIGH
(d) When both inputs are HIGH

Q8. What is the output of a two-input XOR gate when its inputs are different? [Tag: Implementation and validation of logic gates | Easy]

(a) Always unchanged
(b) Always HIGH
(c) Always floating
(d) Always LOW

Q9. Which instrument can be used to check whether a digital signal is HIGH or LOW? [Tag: Implementation and validation of logic gates | Easy]

(a) Logic probe
(b) Solder sucker
(c) Wire stripper
(d) Screwdriver

Q10. Why should unused logic-gate inputs not be left floating during validation? [Tag: Implementation and validation of logic gates | Easy]

(a) They may cause unpredictable outputs
(b) They may increase the wire length
(c) They may change the gate symbol
(d) They may reduce the input count

Q11. Which pair consists entirely of universal logic gates? [Tag: Design of logic gates using universal gates | Easy]

(a) NAND and NOR
(b) AND and OR
(c) OR and XOR
(d) NOT and XOR

Q12. How can a two-input NAND gate be used as a NOT gate for input A ? [Tag: Design of logic gates using universal gates | Easy]

(a) Connect A to both inputs
(b) Connect the output to ground
(c) Leave both inputs unconnected
(d) Connect one input to ground

Q13. Which arrangement implements an AND gate using only NAND gates? [Tag: Design of logic gates using universal gates | Easy]

(a) One NAND gate followed by a NAND inverter
(b) One NAND gate followed by a NOR inverter
(c) One OR gate followed by a NOR inverter
(d) One NOR gate followed by a NAND inverter

Q14. To implement an OR gate using NAND gates, what is done before applying the inputs to the final NAND gate? [Tag: Design of logic gates using universal gates | Easy]

(a) Both inputs are grounded
(b) Both inputs are inverted
(c) Both inputs are disconnected
(d) Both inputs are added

Q15. How can a two-input NOR gate be used as a NOT gate for input A ? [Tag: Design of logic gates using universal gates | Easy]
(a) Leave both inputs floating
(b) Connect A to both inputs
(c) Connect one input to HIGH
(d) Connect the output to HIGH

Q16. Which arrangement implements an OR gate using only NOR gates? [Tag: Design of logic gates using universal gates | Easy]
(a) One AND gate followed by a NAND inverter
(b) One NOR gate followed by a NAND inverter
(c) One NOR gate followed by a NOR inverter
(d) One NAND gate followed by a NOR inverter

Q17. To implement an AND gate using NOR gates, what is done to inputs A and B before the final NOR operation? [Tag: Design of logic gates using universal gates | Easy]
(a) Both inputs are shorted
(b) Both inputs are grounded
(c) Both inputs are inverted
(d) Both inputs are removed

Q18. Which expression represents an OR gate implemented with NAND operations? [Tag: Design of logic gates using universal gates | Easy]
(a) $Y = A \cdot B$
(b) $Y = A + B$
(c) $Y = A \cdot B$
(d) $Y = A \cdot B$

Q19. What does the term "universal gate" mean? [Tag: Design of logic gates using universal gates | Easy]
(a) A gate that always produces a HIGH output
(b) A gate that accepts only one input signal
(c) A gate that can build any Boolean function
(d) A gate that operates without a power supply

Q20. What is the minimum number of two-input NAND gates needed to implement a NOT gate? [Tag: Design of logic gates using universal gates | Easy]
(a) Four gates
(b) Two gates
(c) Three gates
(d) One gate

Q21. During the validation of a two-input XOR gate, which output sequence should be observed for inputs A B = 00 , 01 , 10 , 11 ? [Tag: Implementation and validation of logic gates | Medium]
(a) 0 , 0 , 0 , 1
(b) 1 , 1 , 1 , 0
(c) 0 , 1 , 1 , 0
(d) 1 , 0 , 0 , 1

Q22. A two-input NAND gate produces HIGH for every tested input combination, including A = B = 1 . Which fault best explains this result? [Tag: Implementation and validation of logic gates | Medium]
(a) Both inputs are shorted together
(b) The output is stuck HIGH

(c) The gate behaves as an XOR gate
(d) The output is stuck LOW

Q23. An AND gate is tested with inputs A = 1 and B = 0 , but its output is measured as HIGH. What should the expected output be? [Tag: Implementation and validation of logic gates | Medium]
(a) LOW, because both inputs are not LOW
(b) LOW, because both inputs are not HIGH
(c) HIGH, because one input is HIGH
(d) HIGH, because the inputs are different

Q24. A logic circuit gives the output sequence 1 , 0 , 0 , 1 for inputs A B = 00 , 01 , 10 , 11 . Which gate has been implemented? [Tag: Implementation and validation of logic gates | Medium]
(a) XOR gate
(b) XNOR gate
(c) NAND gate
(d) NOR gate

Q25. While validating a NOT gate, the input is changed from LOW to HIGH. What output transition confirms correct operation? [Tag: Implementation and validation of logic gates | Medium]
(a) HIGH to HIGH
(b) HIGH to LOW
(c) LOW to HIGH
(d) LOW to LOW

Q26. Two switches provide the inputs to an OR gate. An LED connected to the output should turn on under which condition? [Tag: Implementation and validation of logic gates | Medium]
(a) At least one switch is ON
(b) Both switches have equal states
(c) Both switches are OFF
(d) Exactly one switch is OFF

Q27. A two-input NOR gate is operating correctly. For which input combination should its output LED turn on? [Tag: Implementation and validation of logic gates | Medium]
(a) A = 1 , B = 1
(b) A = 1 , B = 0
(c) A = 0 , B = 1
(d) A = 0 , B = 0

Q28. A circuit intended to implement Y = A + B produces LOW only for A = B = 0 . What does this observation indicate? [Tag: Implementation and validation of logic gates | Medium]
(a) The AND gate is functioning correctly
(b) The XOR gate is functioning correctly
(c) The NAND gate is functioning correctly
(d) The OR gate is functioning correctly

Q29. When checking a logic-gate truth table experimentally, why should every possible input combination be applied? [Tag: Implementation and validation of logic gates | Medium]
(a) To verify the complete logical behavior
(b) To increase the output voltage level
(c) To reduce the circuit propagation delay
(d) To determine the package pin count

Q30. An XOR gate has inputs A and B . If B is permanently connected to logic HIGH, how does the output depend on A ? [Tag: Implementation and validation of logic gates | Medium]
(a) The output is always LOW
(b) The output is A
(c) The output is equal to A
(d) The output is always HIGH

Q31. How can a two-input NAND gate be connected to function as a NOT gate? [Tag: Design of logic gates using universal gates | Medium]

(a) Connect the output back to one input
(b) Connect both inputs to the same signal
(c) Connect the two inputs to complementary signals
(d) Connect one input permanently to LOW

Q32. What is the minimum number of two-input NAND gates required to implement a two-input AND gate? [Tag: Design of logic gates using universal gates | Medium]

(a) Three NAND gates
(b) Four NAND gates
(c) Two NAND gates
(d) One NAND gate

Q33. Which NAND-only arrangement implements the OR function $Y = A + B$? [Tag: Design of logic gates using universal gates | Medium]

(a) Invert one input, then NAND both inputs
(b) Invert both inputs, then NAND them
(c) NAND both inputs, then invert the result
(d) NAND both inputs through a single gate

Q34. How can a NOR gate be used as an inverter? [Tag: Design of logic gates using universal gates | Medium]

(a) Apply complementary signals to the inputs
(b) Connect one input permanently to HIGH
(c) Connect both inputs to the same signal
(d) Connect the output to both inputs

Q35. What is the minimum number of two-input NOR gates required to implement a two-input OR gate? [Tag: Design of logic gates using universal gates | Medium]

(a) Two NOR gates
(b) Three NOR gates
(c) Four NOR gates
(d) One NOR gate

Q36. The outputs of two NOR gates are $P = A + B$ and $Q = C + D$. If P and Q are applied to another NOR gate, what is the final expression? [Tag: Design of logic gates using universal gates | Medium]

(a) (A + B) + (C + D)
(b) (A + B) + (C + D)
(c) (A + B) (C + D)
(d) (A + B) (C + D)

Q37. A NAND-only circuit first generates B and then applies A and B to a NAND gate. Which function is produced? [Tag: Design of logic gates using universal gates | Medium]

(a) A + B
(b) A + B
(c) A + B
(d) A + B

Q38. A standard two-input XOR circuit is constructed using only two-input NAND gates. How many NAND gates are required? [Tag: Design of logic gates using universal gates | Medium]

(a) Three NAND gates
(b) Five NAND gates
(c) Two NAND gates
(d) Four NAND gates

Q39. A designer must implement $F = A\,B + C\,D$ using only NAND gates. Which simplification

is most useful? [Tag: Design of logic gates using universal gates | Medium]

(a) F = A BC D
(b) F = A B + C D
(c) F = (A + B) (C + D)
(d) F = A + B + C + D

Q40. Why are NAND and NOR gates described as universal gates? [Tag: Design of logic gates using universal gates | Medium]

(a) Either type can implement any Boolean function
(b) Both types always use only two inputs
(c) Both types produce identical truth tables
(d) Either type eliminates propagation delay

Q41. A NAND-gate implementation is tested with inputs A = 1 and B = 1 . The measured output remains at logic 1 for approximately 40 ns before changing to logic 0. Which conclusion is most appropriate? [Tag: Implementation and validation of logic gates | Hard]

(a) The circuit has an invalid truth table
(b) The circuit has a propagation delay of 40 ns
(c) The circuit implements an XOR gate
(d) The circuit is permanently stuck at logic 1

Q42. Using only 2-input NAND gates, which expression correctly implements the function Y = A + B ? [Tag: Design of logic gates using universal gates | Hard]

(a) Y = (A NAND A) NAND (B NAND B)
(b) Y = (A NAND B) NAND (A NAND B)
(c) Y = A NAND (B NAND B)
(d) Y = (A NAND A) NAND B

Q43. A NOR-based circuit is intended to implement Y = A + B . During validation, the output is measured as logic 1 for (A , B) = (0 , 0) and logic 0 for all other input combinations. What is the best diagnosis? [Tag: Implementation and validation of logic gates | Hard]

(a) The measured behavior matches a NOR gate
(b) The input terminals are electrically shorted
(c) The circuit is behaving as an AND gate
(d) The output should be logic 1 for (1 , 1) only

Q44. What is the minimum number of 2-input NAND gates required to implement Y = A ⊕ B without using any other gate type? [Tag: Design of logic gates using universal gates | Hard]

(a) Four
(b) Three
(c) Six
(d) Five

Q45. A 3-input NAND gate is validated by connecting its unused input permanently to logic 1 while applying signals to the other two inputs. Which 2-input function is observed at the output? [Tag: Implementation and validation of logic gates | Hard]

(a) A + B
(b) A · B
(c) A + B
(d) A · B

Q46. A NOR-only design must implement Y = A · B . Which expression is correct? [Tag: Design of logic gates using universal gates | Hard]

(a) Y = A NOR (B NOR B)
(b) Y = (A NOR B) NOR (A NOR B)
(c) Y = (A NOR A) NOR (B NOR B)
(d) Y = (A NOR A) NOR B

Q47. Two cascaded inverters are used to validate logic-level restoration. The first inverter has t P L H = 18 ns and t P H L = 22 ns ; the second has t P L H = 25 ns and t P H L = 20 ns . What is the maximum rise-or-fall propagation delay through the chain? [Tag: Implementation and validation of logic gates | Hard]

(a) 47 ns
(b) 38 ns
(c) 42 ns
(d) 45 ns

Q48. Using NAND gates, which implementation realizes a 2:1 multiplexer with select input S , data inputs D 0 , D 1 , and output Y = S D 0 + S D 1 ? [Tag: Design of logic gates using universal gates | Hard]

(a) Y = NAND (NAND (S , D 0) , NAND (S , D 1))
(b) Y = NAND (NAND (S , D 0) , NAND (S , D 1))
(c) Y = NAND (NAND (S , D 0) , NAND (S , D 1))
(d) Y = NAND (NAND (S , D 0) , NAND (S , D 1))

Q49. A NAND gate output is connected to an LED through a resistor to ground. The LED turns on when the output is high. Which input sequence tests all logically distinct output transitions most efficiently? [Tag: Implementation and validation of logic gates | Hard]

(a) (1 , 0) → (0 , 1) → (1 , 1) → (0 , 0)
(b) (1 , 1) → (1 , 0) → (0 , 0) → (0 , 1)
(c) (0 , 0) → (0 , 1) → (1 , 1) → (1 , 0)
(d) (0 , 0) → (1 , 0) → (1 , 1) → (0 , 1)

Q50. A NAND-only implementation of Y = A + B + C is required using only 2-input NAND gates. What is the minimum number of gates? [Tag: Design of logic gates using universal gates | Hard]

(a) Five
(b) Seven
(c) Six
(d) Four

Q51. A CMOS NAND input is accidentally left floating during validation. The observed output changes when a nearby jumper wire is moved. Which explanation is most accurate? [Tag: Implementation and validation of logic gates | Hard]

(a) The floating input picks up noise and has an undefined logic level
(b) The output is guaranteed to remain high because CMOS inputs are ideal
(c) The nearby wire changes the Boolean expression of the NAND gate
(d) The NAND gate automatically converts the floating input to logic 0

Q52. A NOR-only circuit implements Y = (A + B) (C + D) . Which structural approach is valid using De Morgan transformations? [Tag: Design of logic gates using universal gates | Hard]

(a) Invert each input, then NOR all four inverted inputs
(b) NOR each pair, then NOR the two pair outputs
(c) NOR all four inputs, then invert only input A
(d) NOR each pair, then OR the two pair outputs directly

Q53. A NAND gate is powered from 5 V . Its input thresholds are V I L (ma x) = 1.5 V and V I H (min) = 3.5 V . A measured input is 2.4 V . How should this input be classified? [Tag: Implementation and validation of logic gates | Hard]

(a) Undefined logic level
(b) Short-circuit current state
(c) Valid logic 1
(d) Valid logic 0

Q54. A function is given by F (A , B , C) = Σ m (1 , 2 , 5 , 6) . Which simplified expression can be implemented efficiently with NAND gates? [Tag: Design of logic gates using universal gates | Hard]

(a) F = A B + A C
(b) F = A B + BC
(c) F = A B + B C
(d) F = B C + B C

Q55. A combinational circuit output briefly glitches when inputs change from 011 to 100 , even though both stable states should produce the same intended output. Which validation method best identifies the cause? [Tag: Implementation and validation of logic gates | Hard]

(a) Measure propagation delays along reconvergent paths
(b) Replace every gate with an inverter
(c) Test only the all-zero input combination
(d) Increase the supply voltage until the glitch disappears

Q56. A NAND-only implementation of Y = A · (B + C) is required. Which expression uses the correct NAND arrangement? [Tag: Design of logic gates using universal gates | Hard]

(a) Y = NAND (NAND (A , B) , NAND (A , C))
(b) Y = NAND (NAND (A , A) , NAND (B , B) NAND NAND (C , C))
(c) Y = NAND (A , NAND (B , C))
(d) Y = NAND (NAND (A , A) , NAND (B , C))

Q57. A 2-input NOR gate is connected with both inputs tied together to signal A . What function should be verified at the output, and what test distinguishes it from an inverter wiring error? [Tag: Implementation and validation of logic gates | Hard]

(a) Y = A ; test both A = 0 and A = 1
(b) Y = A + A ; test both inputs independently
(c) Y = A ; test only A = 1
(d) Y = A + A ; test only A = 0

Q58. Which NAND-only expression implements the 3-input majority function, where the output is 1 when at least two inputs are 1? [Tag: Design of logic gates using universal gates | Hard]

(a) Y = NAND (NAND (A , B) , NAND (A , C))
(b) Y = NAND (NAND (A , B) , NAND (A , C) , NAND (B , C))
(c) Y = NAND (A , NAND (B , C))
(d) Y = NAND (NAND (A , A) , NAND (B , B) , NAND (C , C))

Q59. When validating a gate, an oscilloscope shows the output high level at 4.6 V with no load but only 3.1 V after connecting four gate inputs. Which parameter is most directly implicated? [Tag: Implementation and validation of logic gates | Hard]

(a) Input truth-table ordering
(b) Unused input inversion
(c) Output drive capability and fan-out
(d) Boolean absorption law

Q60. A NOR gate is to be built using NAND gates only. Which sequence correctly preserves the required polarity? [Tag: Design of logic gates using universal gates | Hard]

(a) Invert the NAND output once, then invert both inputs
(b) Invert A and B with NANDs, then NAND the inverted signals
(c) NAND A and B , then connect the result directly to the output
(d) NAND A and B , then invert the NAND output twice

4.2 Section B: Comprehensive Subjective Problem with Detailed Analytical Rubric

Problem 4 **[Workshop Practical / Viva Scenario — 5/10 Marks]**

1 Prove algebraically using De Morgan's theorems that the NAND gate is a universal logic gate. Show complete gate-level realization diagrams and Boolean proofs for: (a) NOT, (b) AND, (c) OR, and (d) XOR gates using only 2-input NAND gates.

Step-by-Step Analytical / Diagnostic Solution:

Universal Logic Proof via NAND Realizations:
A NAND gate produces $Y = \overline{A \cdot B}$.

(a) NOT Gate Realization (1 NAND Gate):
Tie both inputs together: $B = A$.

$$Y = \overline{A \cdot A} = \overline{A}$$

(b) AND Gate Realization (2 NAND Gates):
Invert the output of a NAND gate using a second NAND configured as an inverter:

$$Y_1 = \overline{A \cdot B}, \quad Y = \overline{Y_1 \cdot Y_1} = \overline{\overline{A \cdot B}} = A \cdot B$$

(c) OR Gate Realization (3 NAND Gates):
Invert inputs A and B individually, then feed into a third NAND gate:

$$Y_1 = \overline{A}, \quad Y_2 = \overline{B}$$

$$Y = \overline{Y_1 \cdot Y_2} = \overline{\overline{A} \cdot \overline{B}}$$

Applying De Morgan's theorem $\overline{X \cdot Y} = \overline{X} + \overline{Y}$:

$$Y = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$

(d) XOR Gate Realization (4 NAND Gates):
Boolean identity for XOR: $A \oplus B = \overline{A}B + A\overline{B} = A\overline{A \cdot B} + B\overline{A \cdot B}$.
Let $G_1 = \overline{A \cdot B}$.
Feed A and G_1 into G_2: $Y_2 = \overline{A \cdot G_1} = \overline{A \cdot \overline{AB}}$.
Feed B and G_1 into G_3: $Y_3 = \overline{B \cdot G_1} = \overline{B \cdot \overline{AB}}$.
Feed Y_2 and Y_3 into G_4:

$$Y = \overline{Y_2 \cdot Y_3} = \overline{\overline{A\overline{AB}} \cdot \overline{B\overline{AB}}} = A\overline{AB} + B\overline{AB} = A(\overline{A} + \overline{B}) + B(\overline{A} + \overline{B}) = A\overline{B} + \overline{A}B = A \oplus B$$

Examiner Rubric (10 Marks):
- Statement and explanation of universality definition: 2 Marks
- Inverter and AND realizations with proofs: 2 Marks
- OR realization with De Morgan's theorem step: 3 Marks
- 4-NAND XOR realization with algebraic derivation: 3 Marks

4.3 Section C: Laboratory Viva-Voce Questions with Authoritative Solutions

Viva Question: What is the difference between TTL and CMOS digital ICs regarding unused inputs?
Technical Answer: In standard TTL (74LS series), an open/floating input behaves as a logic HIGH due to

internal emitter pull-up, though it acts as an antenna susceptible to noise. In CMOS (74HC/CD4000 series), input impedance is $> 10^{12}\,\Omega$; floating inputs drift unpredictably between logic thresholds, causing excessive shoot-through current and oscillation. All unused CMOS inputs MUST be tied to V_{CC} or GND.

Viva Question: What is the propagation delay of a logic gate and how is it measured on a DSO?
Technical Answer: Propagation delay is the finite time required for an output transition to occur after an input stimulus. It is measured on a DSO between the 50% amplitude point of the input waveform and the 50% amplitude point of the resulting output waveform (t_{pLH} and t_{pHL}).

Viva Question: Why are NAND and NOR gates preferred in semiconductor fabrication over AND and OR gates?
Technical Answer: In CMOS technology, an inverting gate (NAND or NOR) requires only 4 transistors. Non-inverting gates (AND, OR) require an internal inverting stage, requiring 6 transistors. Hence, NAND/NOR gates are smaller, faster, and consume less dynamic power.

4.4 Section D: Workshop Troubleshooting & Fault Diagnostic Case Study

Troubleshooting Scenario: Floating Inputs Produce Random Output States
A student builds an AND gate circuit using a 74LS08 IC. One input is wired to a toggle switch, while the second input is left unconnected on the breadboard. The output LED turns ON when the switch is HIGH and OFF when LOW, behaving like a buffer.

Diagnosis: The ungrounded TTL input floated to logic HIGH ($V >$ 2.0
$textV$). Consequently, $Y = A$
$cdot1 = A$. However, touching the breadboard or bringing a hand near the IC causes erratic flickering due to static capacitive pickup.
Fix: Always pull unused logic inputs explicitly to V_{CC} (through a 1
,
$mathrmk$
$Omega$ pull-up resistor) or ground.

Chapter 5

Unit 5: Combinational Arithmetic & Logic Circuits: Half/Full Adders & Subtractors

5.1 Section A: Multiple Choice Practice Questions (60 Curated Problems)

The following 60 multiple-choice questions reflect the official workshop evaluation bank, categorized by fundamental learning outcome and laboratory testing difficulty.

Q1. How many input bits are used in a half adder? [Tag: Design of a half adder | Easy]
(a) Two input bits
(b) Three input bits
(c) One input bit
(d) Four input bits

Q2. What are the two outputs of a half adder? [Tag: Design of a half adder | Easy]
(a) Input and output
(b) Enable and select
(c) Clock and reset
(d) Sum and carry

Q3. Which logic gate produces the Sum output of a half adder? [Tag: Design of a half adder | Easy]
(a) NOT gate
(b) XOR gate
(c) AND gate
(d) OR gate

Q4. Which logic gate produces the Carry output of a half adder? [Tag: Design of a half adder | Easy]
(a) OR gate
(b) AND gate
(c) XOR gate
(d) NOT gate

Q5. What is the Sum output of a half adder when A = 0 and B = 0 ? [Tag: Design of a half adder | Easy]
(a) 2
(b) Undefined
(c) 1
(d) 0

Q6. What is the Carry output of a half adder when A = 1 and B = 1 ? [Tag: Design of a half adder |

Easy]

(a) Undefined
(b) 1
(c) 2
(d) 0

Q7. Which operation does a half adder perform? [Tag: Design of a half adder | Easy]

(a) Bit comparison
(b) Decimal multiplication
(c) Binary subtraction
(d) Single-bit addition

Q8. How many input bits are used in a full adder? [Tag: Design of a full adder | Easy]

(a) Three input bits
(b) Two input bits
(c) One input bit
(d) Four input bits

Q9. What is the additional input of a full adder compared with a half adder? [Tag: Design of a full adder | Easy]

(a) Clock signal
(b) Input carry
(c) Output carry
(d) Output sum

Q10. What are the outputs of a full adder? [Tag: Design of a full adder | Easy]

(a) Clock and sum
(b) Select and enable
(c) Sum and carry
(d) Input and carry

Q11. Which expression represents the Sum output of a full adder? [Tag: Design of a full adder | Easy]

(a) A + B + C in
(b) A B C in
(c) A · B · C in
(d) A ⊕ B ⊕ C in

Q12. What is the Sum output of a full adder when A = 1 , B = 0 , and C in = 0 ? [Tag: Design of a full adder | Easy]

(a) 2
(b) 0
(c) 1
(d) Undefined

Q13. What is the Carry output of a full adder when A = 1 , B = 1 , and C in = 0 ? [Tag: Design of a full adder | Easy]

(a) 1
(b) 2
(c) Undefined
(d) 0

Q14. A full adder can be constructed using how many half adders and one OR gate? [Tag: Design of a full adder | Easy]

(a) One half adder
(b) Three half adders
(c) Two half adders
(d) Four half adders

Q15. What does C in represent in a full adder? [Tag: Design of a full adder | Easy]

(a) Input clock
(b) Input carry
(c) Output control
(d) Output carry

Q16. Which tool is commonly used to list all input and output combinations of a logic circuit? [Tag: Validation of combinational logic | Easy]

(a) Timing clock
(b) Connection chart
(c) Power table
(d) Truth table

Q17. What should be compared when validating a combinational logic circuit? [Tag: Validation of combinational logic | Easy]

(a) Gate sizes and labels
(b) Expected and observed outputs
(c) Input and wire colors
(d) Power and ground lengths

Q18. How many input combinations are possible for a circuit with three binary inputs? [Tag: Validation of combinational logic | Easy]

(a) Eight combinations
(b) Six combinations
(c) Nine combinations
(d) Three combinations

Q19. What is a combinational logic circuit's output based on? [Tag: Validation of combinational logic | Easy]

(a) Previous output values
(b) Clock frequency values
(c) Present input values
(d) Stored memory values

Q20. Which method can be used to validate an adder circuit in a workshop? [Tag: Validation of combinational logic | Easy]

(a) Measuring wire length
(b) Changing the circuit color
(c) Removing all input signals
(d) Testing input combinations

Q21. A half adder receives inputs A = 1 and B = 1 . What are its sum and carry outputs? [Tag: Design of a half adder | Medium]

(a) Sum = 1, Carry = 1
(b) Sum = 1, Carry = 0
(c) Sum = 0, Carry = 0
(d) Sum = 0, Carry = 1

Q22. Which combination of logic gates correctly implements a half adder? [Tag: Design of a half adder | Medium]

(a) AND gate for sum and XOR gate for carry
(b) OR gate for sum and AND gate for carry
(c) NOR gate for sum and XOR gate for carry
(d) XOR gate for sum and AND gate for carry

Q23. A half adder has a sum output of 1 and a carry output of 0 . Which input condition must be present? [Tag: Design of a half adder | Medium]

(a) Both B and C are possible

(b) A = 0 , B = 1
(c) A = 1 , B = 0
(d) A = 0 , B = 0

Q24. If the AND gate in a half-adder circuit is replaced by an OR gate while the XOR gate remains unchanged, which output behavior is affected? [Tag: Design of a half adder | Medium]

(a) Both outputs become complements
(b) Both outputs remain unchanged
(c) Only the carry output is affected
(d) Only the sum output is affected

Q25. Which input combination produces a carry but no sum in a half adder? [Tag: Design of a half adder | Medium]

(a) A = 1 , B = 0
(b) A = 1 , B = 1
(c) A = 0 , B = 1
(d) A = 0 , B = 0

Q26. A half adder is used to add two single-bit numbers. For inputs A = 1 and B = 0 , what binary value is represented by the combined carry and sum outputs? [Tag: Design of a half adder | Medium]

(a) 0 0 2
(b) 1 1 2
(c) 0 1 2
(d) 1 0 2

Q27. A designer wants to implement the half-adder sum using only basic AND, OR, and NOT gates. Which Boolean expression should be used? [Tag: Design of a half adder | Medium]

(a) A + B
(b) A B + A B
(c) A B + A B
(d) A B

Q28. A full adder receives A = 1 , B = 0 , and carry-in C in = 1 . What are the sum and carry-out? [Tag: Design of a full adder | Medium]

(a) Sum = 1, Carry-out = 0
(b) Sum = 1, Carry-out = 1
(c) Sum = 0, Carry-out = 0
(d) Sum = 0, Carry-out = 1

Q29. Which Boolean expression represents the sum output of a full adder? [Tag: Design of a full adder | Medium]

(a) A B + A C in + B C in
(b) A + B + C in
(c) A ⊕ B ⊕ C in
(d) A ⊕ B ⊕ C in

Q30. Which expression correctly represents the carry-out of a full adder? [Tag: Design of a full adder | Medium]

(a) A B + A C in + B C in
(b) A + B + C in
(c) A ⊕ B ⊕ C in
(d) A B + C in

Q31. How can a full adder be constructed using half adders and an OR gate? [Tag: Design of a full adder | Medium]

(a) Two half adders and one AND gate
(b) Two half adders and one OR gate

(c) One half adder and one OR gate
(d) Three half adders and one XOR gate

Q32. For a full adder, the sum output is 1 and the carry-out is 1 . Which input condition is possible? [Tag: Design of a full adder | Medium]

(a) A = 1 , B = 1 , C in = 0
(b) A = 0 , B = 1 , C in = 0
(c) A = 0 , B = 0 , C in = 1
(d) A = 1 , B = 1 , C in = 1

Q33. A full adder has inputs A = 0 , B = 1 , and C in = 0 . What is the decimal value represented by its outputs? [Tag: Design of a full adder | Medium]

(a) Zero
(b) Two
(c) Three
(d) One

Q34. A full-adder circuit is designed with two XOR gates, two AND gates, and one OR gate. Which signal is normally connected to the second XOR gate along with the first XOR output? [Tag: Design of a full adder | Medium]

(a) Carry-out C o u t
(b) Input A
(c) Input B
(d) Carry-in C in

Q35. What is the most appropriate method for validating a newly built full-adder circuit? [Tag: Validation of combinational logic | Medium]

(a) Compare measured outputs with the complete truth table
(b) Test only the all-zero input
(c) Measure only the supply voltage
(d) Check the circuit only when the sum is one

Q36. During testing of a half adder, the carry output is always 0 , including for A = 1 and B = 1 . Which fault is most likely? [Tag: Validation of combinational logic | Medium]

(a) The output indicator is connected to ground
(b) The supply frequency is too high
(c) The AND gate or its wiring is faulty
(d) The XOR input is shorted

Q37. A full-adder test gives the correct sum for all input combinations but an incorrect carry-out. Which part should be checked first? [Tag: Validation of combinational logic | Medium]

(a) The carry-generation network
(b) The circuit's clock source
(c) The input switches for A only
(d) The XOR network for the sum

Q38. When validating combinational logic on a breadboard, why should unused logic inputs not be left floating? [Tag: Validation of combinational logic | Medium]

(a) Floating inputs always force the output to zero
(b) Floating inputs can produce unpredictable logic levels
(c) Floating inputs increase the circuit's clock speed
(d) Floating inputs permanently damage every logic gate

Q39. A measured full-adder output disagrees with the truth table only when A = 1 , B = 1 , and C in = 1 . What output should be expected for this test? [Tag: Validation of combinational logic | Medium]

(a) Sum = 0, Carry-out = 0
(b) Sum = 1, Carry-out = 0

(c) Sum = 1, Carry-out = 1
(d) Sum = 0, Carry-out = 1

Q40. A logic probe shows an undefined reading at the sum output while testing a half adder. Which action is most suitable before concluding that the design is incorrect? [Tag: Validation of combinational logic | Medium]

(a) Ignore the reading and test only carry
(b) Verify power, ground, and input connections
(c) Replace the XOR gate with a NOT gate
(d) Tie the sum output directly to supply

Q41. A half adder is implemented using only NAND gates. Which minimized two-output realization correctly produces sum S and carry C for inputs A and B ? [Tag: Design of a half adder | Hard]

(a) S = (A N A N D A) N A N D (B N A N D B) , C = (A N A N D B) N A N D (A N A N D B)
(b) S = (A N A N D B) N A N D ((A N A N D (A N A N D B)) N A N D (B N A N D (A N A N D B))) , C = (A N A N D B) N A N D (A N A N D B)
(c) S = (A N A N D B) N A N D ((A N A N D (A N A N D B)) N A N D (B N A N D (A N A N D B))) , C = A N A N D B
(d) S = (A N A N D B) N A N D (A N A N D B) , C = (A N A N D A) N A N D (B N A N D B)

Q42. A half adder uses active-high inputs and outputs. Its sum output is accidentally connected to an AND gate while its carry output is connected to an XOR gate. Which input condition exposes both wiring errors simultaneously? [Tag: Design of a half adder | Hard]

(a) A = 0 , B = 0
(b) A = 0 , B = 1
(c) A = 1 , B = 0
(d) A = 1 , B = 1

Q43. A half-adder sum is specified as S = A $\oplus$ B , but the available gates are one 2-input OR gate and one 2-input AND gate. An inverter is also available. Which expression implements the sum with the fewest additional gates? [Tag: Design of a half adder | Hard]

(a) S = (A + B) A + B
(b) S = (A + B) A B
(c) S = A + B + A B
(d) S = A B + A + B

Q44. A half adder is described by the canonical sum-of-products expression for its carry output. Which expression is correct, and why is no minterm required for the sum output at input A B = 11 ? [Tag: Design of a half adder | Hard]

(a) C = Σ m (3) ; the sum is one when both inputs are one
(b) C = Σ m (0 , 3) ; the sum is zero when both inputs are equal
(c) C = Σ m (2 , 3) ; the sum is one whenever A is one
(d) C = Σ m (1 , 2) ; the sum is zero when both inputs are one

Q45. Two half adders are cascaded incorrectly by feeding the first half-adder carry into the second half-adder along with an external bit X . For A = 1 , B = 1 , and X = 1 , what are the final sum and carry outputs of this cascade? [Tag: Design of a half adder | Hard]

(a) S = 1 , C = 1
(b) S = 1 , C = 0
(c) S = 0 , C = 1
(d) S = 0 , C = 0

Q46. A half-adder circuit uses positive-logic XOR and AND gates, but its inputs are active-low signals A L and B L . Which output equations represent active-high versions of the arithmetic sum and carry? [Tag: Design of a half adder | Hard]

(a) S = A L $\oplus$ B L , C = A L B L
(b) S = A L $\oplus$ B L , C = A L B L

(c) $S = A\ L \oplus B\ L$, $C = A\ L + B\ L$
(d) $S = A\ L + B\ L$, $C = A\ L\ B\ L$

Q47. A half-adder implementation has propagation delays of 8 ns through its XOR path and 5 ns through its AND path. If both inputs change simultaneously from 00 to 11 , which transient behavior is expected under an inertial-delay model? [Tag: Design of a half adder | Hard]

(a) The sum remains low and the carry rises after 5 ns
(b) The sum rises after 8 ns and the carry remains low
(c) The sum pulses high before returning low, while carry rises
(d) The sum and carry rise together after the XOR delay

Q48. Which Boolean expression gives the carry output of a full adder in terms of augend A , addend B , and carry-in C in ? [Tag: Design of a full adder | Hard]

(a) $C\ o\ u\ t = A\ B + C\ in\ (\ A \oplus B\ )$
(b) $C\ o\ u\ t = A\ B + A\ C\ in + B\ C\ in$
(c) $C\ o\ u\ t = A \oplus B \oplus C\ in$
(d) $C\ o\ u\ t = (\ A + B\ )\ C\ in + A\ B$

Q49. A full adder is constructed from two half adders and an OR gate. If the first half adder produces (S 1 , C 1) from A , B and the second produces (S , C 2) from S 1 , C in , which carry equation results? [Tag: Design of a full adder | Hard]

(a) $C\ o\ u\ t = C\ 1 + C\ 2 = A + B + C\ in$
(b) $C\ o\ u\ t = C\ 1\ C\ 2 = A\ B\ C\ in$
(c) $C\ o\ u\ t = C\ 1 \oplus C\ 2 = A\ B \oplus C\ in$
(d) $C\ o\ u\ t = C\ 1 + C\ 2 = A\ B + C\ in\ (\ A \oplus B\ )$

Q50. For a full adder with inputs A = 1 , B = 0 , and C in = 1 , which output pair is correct and what arithmetic condition does it represent? [Tag: Design of a full adder | Hard]

(a) $S = 1$, $C\ o\ u\ t = 1$; the total is binary 11
(b) $S = 0$, $C\ o\ u\ t = 1$; the total is binary 10
(c) $S = 1$, $C\ o\ u\ t = 0$; the total is binary 01
(d) $S = 0$, $C\ o\ u\ t = 0$; the total is binary 00

Q51. A full-adder sum output is implemented as S = (A ⊕ B) ⊕ C in . If the first XOR gate has a stuck-at-zero output, for which input set is the observed sum incorrect? [Tag: Design of a full adder | Hard]

(a) $A = 0$, $B = 1$, $C\ in = 1$
(b) $A = 1$, $B = 1$, $C\ in = 1$
(c) $A = 0$, $B = 0$, $C\ in = 0$
(d) $A = 1$, $B = 0$, $C\ in = 0$

Q52. A ripple-carry adder is formed from four full adders. The carry propagation delay of each stage is 12 ns , and the final sum logic settles 7 ns after its carry-in settles. What is the worst-case delay from a carry-in change at the least significant stage to the most significant sum output? [Tag: Design of a full adder | Hard]

(a) 43 ns
(b) 48 ns
(c) 36 ns
(d) 19 ns

Q53. A full adder must be designed using only 2-input XOR, AND, and OR gates. Which gate-count combination implements both outputs with no shared intermediate node? [Tag: Design of a full adder | Hard]

(a) Two XOR, three AND, one OR
(b) Two XOR, two AND, one OR
(c) One XOR, two AND, two OR
(d) Three XOR, two AND, two OR

Q54. For a full adder, the carry output is required to be independent of C in for a particular pair

(A , B) . Which pair satisfies this condition, and what is the resulting carry value? [Tag: Design of a full adder | Hard]

(a) (1 , 0) ; C o u t = 1 for either carry-in
(b) (0 , 0) ; C o u t = 0 for either carry-in
(c) (0 , 1) ; C o u t = 0 for either carry-in
(d) (1 , 1) ; C o u t = C in for either carry-in

Q55. A full adder is used to add three one-bit operands. If the sum output is correct but the carry output is replaced by C o u t ' = A B + A C in , which input combination produces an incorrect carry? [Tag: Design of a full adder | Hard]

(a) A = 1 , B = 0 , C in = 1
(b) A = 0 , B = 0 , C in = 1
(c) A = 0 , B = 1 , C in = 1
(d) A = 1 , B = 1 , C in = 0

Q56. A combinational circuit has three inputs, and its output is one for minterms m (1 , 2 , 5 , 7) . Which input sequence provides complete functional validation when all possible static input combinations are tested? [Tag: Validation of combinational logic | Hard]

(a) 111 , 110 , 101 , 100 , 011 , 010 , 001 , 000
(b) 000 , 011 , 100 , 110
(c) 000 , 001 , 010 , 011 , 100 , 101 , 110 , 111
(d) 001 , 010 , 101 , 111

Q57. A measured full-adder truth table shows the correct outputs for all cases except A = 1 , B = 0 , C in = 1 , where the measured result is S = 1 , C o u t = 0 . Which fault best explains this isolated error? [Tag: Validation of combinational logic | Hard]

(a) The input B is permanently held at one
(b) The sum XOR network is stuck at zero
(c) The carry output is stuck at zero
(d) The carry term A C in is missing

Q58. During validation of a two-level combinational circuit, the output briefly pulses when input A changes from 0 to 1 while B = 1 . The steady-state function is F = A + B . Which explanation is most appropriate? [Tag: Validation of combinational logic | Hard]

(a) The circuit has a static-1 hazard caused by unequal path delays
(b) The truth table is invalid because OR gates cannot produce pulses
(c) The circuit is sequential because the output depends on prior inputs
(d) The circuit has a static-0 hazard because both inputs are zero

Q59. A logic analyzer samples a full-adder output at the exact instant inputs change. Why can this method falsely suggest an incorrect truth-table entry? [Tag: Validation of combinational logic | Hard]

(a) Truth tables apply only when inputs change asynchronously
(b) Logic analyzers automatically invert carry outputs
(c) Combinational outputs may be transient before propagation settles
(d) Full adders retain the previous output until reset

Q60. A half-adder testbench checks only input pairs 00 and 11 and observes correct outputs. Which conclusion is justified? [Tag: Validation of combinational logic | Hard]

(a) Both functions are fully validated because equal inputs were tested
(b) The sum function is validated, but the carry function is not validated
(c) The carry function is validated, but the sum function is not fully validated
(d) Neither function is validated because unequal inputs were omitted

5.2 Section B: Comprehensive Subjective Problem with Detailed Analytical Rubric

Problem 5 **[Workshop Practical / Viva Scenario — 5/10 Marks]**

1 (a) Formulate the truth table for a 1-bit Full Adder with inputs A, B, C_{in} and outputs Sum(S) and Carry(C_{out}). (b) Derive the minimal Boolean expressions for S and C_{out}. (c) Draw the block-level logic realization using two Half Adders and one OR gate, and prove algebraic equivalence.

Step-by-Step Analytical / Diagnostic Solution:

(a) Full Adder Truth Table:

A&B&C_{in}&Sum (S)&C_{out}
0&0&0&0&0
0&0&1&1&0
0&1&0&1&0
0&1&1&0&1
1&0&0&1&0
1&0&1&0&1
1&1&0&0&1
1&1&1&1&1

(b) Minimal Boolean Expressions:
From the minterms of Sum:

$$S = \sum m(1,2,4,7) = \overline{A}\overline{B}C_{\text{in}} + \overline{A}B\overline{C_{\text{in}}} + A\overline{B}\overline{C_{\text{in}}} + ABC_{\text{in}}$$

Factoring:

$$S = C_{\text{in}}(\overline{A}\overline{B} + AB) + \overline{C_{\text{in}}}(\overline{A}B + A\overline{B}) = C_{\text{in}}\overline{(A \oplus B)} + \overline{C_{\text{in}}}(A \oplus B) = A \oplus B \oplus C_{\text{in}}$$

From the minterms of Carry:

$$C_{\text{out}} = \sum m(3,5,6,7) = AB + BC_{\text{in}} + AC_{\text{in}} = AB + C_{\text{in}}(A + B) = AB + C_{\text{in}}(A \oplus B)$$

(c) Implementation Using Two Half Adders and OR Gate:
Half Adder 1: Inputs $A, B \implies S_1 = A \oplus B$, $C_1 = AB$.
Half Adder 2: Inputs $S_1, C_{\text{in}} \implies S = S_1 \oplus C_{\text{in}} = (A \oplus B) \oplus C_{\text{in}}$, $C_2 = S_1 \cdot C_{\text{in}} = (A \oplus B)C_{\text{in}}$.
The final carry is formed by an OR gate: $C_{\text{out}} = C_1 + C_2 = AB + (A \oplus B)C_{\text{in}}$.
This matches the simplified Full Adder carry expression identically.

Examiner Rubric (10 Marks):
- Complete and accurate truth table: 2 Marks
- Algebraic extraction and simplification of Sum $S = A \oplus B \oplus C_{\text{in}}$: 3 Marks
- Minimal derivation of Carry $C_{\text{out}} = AB + C_{\text{in}}(A \oplus B)$: 3 Marks
- Two Half-Adder block diagram and cascading proof: 2 Marks

5.3 Section C: Laboratory Viva-Voce Questions with Authoritative Solutions

Viva Question: Why cannot a Half Adder alone be used to cascade multi-bit binary addition?
Technical Answer: A Half Adder only accepts two single-bit inputs (A and B). In multi-bit addition, every stage except the least significant bit (LSB) must also incorporate an incoming carry (C_{in}) generated by the preceding bit stage. Therefore, Full Adders are required.

Viva Question: What is a ripple-carry adder and what is its primary performance limitation?
Technical Answer: A ripple-carry adder connects n Full Adders in series, where each stage passes its carry-out to the next stage's carry-in. Its limitation is propagation delay: the most significant bit (MSB) cannot produce its valid sum until the carry ripples through all preceding stages ($t_{\text{total}} = n \cdot t_{\text{carry}}$).

Viva Question: How can a 4-bit binary adder (IC 7483) be converted into an Adder/Subtractor?
Technical Answer: Connect the B inputs to the outputs of four XOR gates. The second input of each XOR gate is connected to an Add/$\overline{\text{Sub}}$ control line, which is also fed into the adder's initial carry input C_0. When Mode $= 0$, B passes un-inverted and $C_0 = 0$ (Addition). When Mode $= 1$, B is inverted (1's complement) and $C_0 = 1$ adds 1, performing 2's complement subtraction: $A + \overline{B} + 1 = A - B$.

5.4 Section D: Workshop Troubleshooting & Fault Diagnostic Case Study

Troubleshooting Scenario: Sum Output Inverted on Breadboard
A student wires a Full Adder circuit using a 7486 XOR IC and a 7408 AND IC. Testing with inputs $A = 1, B = 1, C_{\text{in}} = 0$ gives $S = 1$ and $C_{\text{out}} = 0$, which is the exact inverse of the true output ($S = 0, C_{\text{out}} = 1$).

Diagnosis: Pin miswiring on the 7486 quad XOR chip. The student connected the output probe to Pin 4 (an input to gate 2) instead of Pin 3 (the output of gate 1), leaving the sum line driven by an undefined high-impedance state.
Fix: Verify IC orientation notch and match schematic node labels with absolute DIP pin numbers before powering the trainer.

Chapter 6

Unit 6: Microcontroller Workshop: Arduino Uno & Embedded Sensor Interfacing

6.1 Section A: Multiple Choice Practice Questions (60 Curated Problems)

The following 60 multiple-choice questions reflect the official workshop evaluation bank, categorized by fundamental learning outcome and laboratory testing difficulty.

Q1. What is the Arduino Uno? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) A type of battery charger
(b) A digital storage device
(c) A wireless router
(d) A microcontroller development board

Q2. Which microcontroller is used on the Arduino Uno board? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) ESP8266
(b) ATmega328P
(c) ATmega2560
(d) PIC16F877A

Q3. What does IR stand for in an IR sensor? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) Input Relay
(b) Infrared Radiation
(c) Internal Resistance
(d) Integrated Register

Q4. What is the main purpose of an IR sensor in a detection system? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) To detect nearby objects
(b) To regulate voltage
(c) To produce sound
(d) To store the program

Q5. Which Arduino Uno pin is commonly used to provide ground? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) VIN pin
(b) GND pin

(c) AREF pin
(d) RESET pin

Q6. Which Arduino function runs once when the board starts? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) start()
(b) loop()
(c) begin()
(d) setup()

Q7. Which Arduino function repeats continuously after setup() ? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) cycle()
(b) runOnce()
(c) repeat()
(d) loop()

Q8. Which function configures an Arduino pin as an input or output? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) digitalRead()
(b) analogRead()
(c) digitalWrite()
(d) pinMode()

Q9. Which function reads a digital signal from an IR sensor? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) digitalWrite()
(b) analogWrite()
(c) digitalRead()
(d) pinMode()

Q10. Which function sends a HIGH or LOW signal to a digital output pin? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) analogRead()
(b) pinMode()
(c) digitalRead()
(d) digitalWrite()

Q11. What type of signal does a digital IR sensor module commonly provide? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) Audio or video
(b) HIGH or LOW
(c) Positive or negative voltage only
(d) Text or image data

Q12. Which component on many IR sensor modules emits infrared light? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) Phototransistor
(b) Voltage regulator
(c) Buzzer
(d) IR LED

Q13. Which component receives reflected infrared light? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]
(a) Crystal oscillator
(b) Push button
(c) Photodiode
(d) Relay coil

Q14. What is the purpose of the potentiometer on many IR sensor modules? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) To adjust detection sensitivity
(b) To store sensor readings
(c) To increase program memory
(d) To connect the USB cable

Q15. Which Arduino Uno pin group is intended for digital input and output? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) Digital pins
(b) Analog reference pins
(c) Power pins
(d) USB shield pins

Q16. What is the usual logic value of a HIGH digital signal? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) Logic 1
(b) A stored program
(c) Logic 0
(d) No connection

Q17. Which software is commonly used to write and upload programs to an Arduino Uno? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) Spreadsheet editor
(b) Web browser
(c) Image viewer
(d) Arduino IDE

Q18. What is an Arduino program commonly called? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) A waveform
(b) A register
(c) A sketch
(d) A terminal

Q19. Why is a common ground needed between the Arduino and IR sensor? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) To replace the power supply
(b) To provide a shared reference
(c) To erase the Arduino program
(d) To increase infrared brightness

Q20. What can an Arduino do when an object is detected by an IR sensor? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Easy]

(a) Increase the object size
(b) Change its processor
(c) Recharge the sensor battery
(d) Turn on an LED

Q21. An IR sensor module gives a LOW output when an object is detected. Which Arduino Uno code correctly turns on an LED connected to pin 13 when the object is present? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) if (analogRead(sensorPin) == LOW) digitalWrite(13, HIGH);
(b) if (digitalRead(sensorPin) == LOW) digitalWrite(13, HIGH);
(c) if (digitalWrite(sensorPin, LOW)) digitalRead(13, HIGH);
(d) if (digitalRead(sensorPin) == HIGH) digitalWrite(13, HIGH);

Q22. Why is a common ground connection required between an Arduino Uno and an IR sensor

module? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) It converts the sensor output into analog voltage
(b) It prevents the Arduino from requiring a power supply
(c) It provides a shared voltage reference for signals
(d) It increases the sensor's infrared wavelength

Q23. An IR obstacle sensor is connected to digital pin 7, and its output is active HIGH. Which initialization is appropriate? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) pinMode(7, INPUT);
(b) digitalWrite(7, INPUT);
(c) pinMode(7, OUTPUT);
(d) analogWrite(7, INPUT);

Q24. The detection range of an IR module changes when its onboard potentiometer is adjusted. What does this potentiometer usually control? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) The comparator's detection threshold
(b) The Arduino clock frequency
(c) The infrared LED color
(d) The serial communication speed

Q25. An LED is connected directly between an Arduino output pin and ground. Why should a series resistor be included? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) To reverse the LED's polarity
(b) To convert digital output into serial data
(c) To limit current through the LED
(d) To increase the Arduino supply voltage

Q26. An IR sensor output repeatedly changes between HIGH and LOW when an object is stationary near the detection boundary. Which software method is most suitable for reducing false switching? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) Require a stable state for several readings
(b) Set every unused pin to analog input
(c) Increase the serial monitor baud rate
(d) Change the LED resistor to a larger value

Q27. Which statement best describes the role of the IR receiver in a typical reflective obstacle sensor? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) It detects reflected infrared radiation
(b) It stores the detected object's distance
(c) It regulates current to the indicator LED
(d) It generates the Arduino program clock

Q28. A sensor module operates at 5 V, while its output is connected to Arduino Uno digital pin 8. Which connection is correct? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) VCC to GND, GND to 5 V, OUT to pin 8
(b) VCC to Vin, GND to pin 8, OUT to the USB shield
(c) VCC to 5 V, GND to GND, OUT to pin 8
(d) VCC to pin 8, GND to 5 V, OUT to GND

Q29. What is the main purpose of using Serial.begin(9600) in an IR detection project? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) To set the IR transmitter's emission frequency
(b) To select the sensor's physical detection range

(c) To provide regulated power to the sensor module
(d) To configure communication with the Serial Monitor

Q30. A detection system should keep an alarm active while an object remains in front of the sensor. Which control structure is most appropriate? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) Use analogWrite without checking the sensor
(b) Use a one-time statement in setup only
(c) Use an if-else statement based on sensor state
(d) Use delay without reading the sensor again

Q31. Why can a black object be detected less reliably by a reflective IR sensor than a white object? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) Black objects disable the Arduino input circuit
(b) Black surfaces always produce a higher supply voltage
(c) Black objects emit visible light instead of infrared light
(d) Black surfaces generally reflect less infrared light

Q32. An IR sensor is connected to pin 2, and an indicator LED is connected to pin 9. Which pair of declarations matches this arrangement? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) const int sensorPin = 0; const int ledPin = 1;
(b) const int sensorPin = A2; const int ledPin = A9;
(c) const int sensorPin = 2; const int ledPin = 9;
(d) const int sensorPin = 9; const int ledPin = 2;

Q33. What is the likely result if an IR sensor's OUT terminal is accidentally connected to an Arduino output pin configured as OUTPUT? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) The Arduino increases the sensor detection range
(b) The IR receiver stops needing a common ground
(c) The sensor and Arduino outputs may conflict
(d) The sensor automatically becomes an analog input

Q34. An IR sensor indicates detection with LOW, but the program activates the alarm with HIGH. Which change fixes the logical error? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) Move the sensor wire to the USB connector
(b) Invert the condition used to test the sensor
(c) Replace digitalRead with analogWrite
(d) Increase the delay time without changing the condition

Q35. Why should an IR sensor and its target generally be kept aligned during testing? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) Alignment changes the USB communication protocol
(b) Alignment determines the Arduino's program memory size
(c) Alignment controls the resistor tolerance automatically
(d) Alignment affects the amount of reflected IR received

Q36. Which sequence represents a suitable Arduino program structure for an IR detection system? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]
(a) Read sensor once, configure pins, stop execution
(b) Configure pins, read sensor, control output, repeat
(c) Start serial output, remove power, configure pins, stop
(d) Control output, erase program, read sensor, repeat

Q37. A sensor output is connected to an Arduino input, but the reading changes randomly when the sensor is disconnected. What is the most likely cause? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) The Arduino has too many comments
(b) The LED has excessive brightness
(c) The input pin is floating
(d) The USB cable is too short

Q38. An Arduino Uno must count an object each time it passes the IR sensor, rather than counting repeatedly while it remains present. What programming technique is needed? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) Place the counter declaration inside setup only
(b) Detect a transition between absent and present states
(c) Use a constant HIGH output for the sensor pin
(d) Keep increasing the counter every loop cycle

Q39. Which change can help prevent an Arduino output pin from being overloaded when driving an external buzzer circuit? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) Use a transistor driver and suitable protection
(b) Short the buzzer terminals to the 5 V rail
(c) Connect the buzzer directly to the input pin
(d) Remove the common ground from the circuit

Q40. An IR detection circuit works indoors but gives false detections under strong sunlight. What is a practical improvement? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Medium]

(a) Replace digitalRead with pinMode
(b) Increase the Arduino reset frequency
(c) Shield the sensor from ambient infrared light
(d) Disconnect the sensor ground during operation

Q41. An IR obstacle sensor is connected to Arduino Uno digital pin D2 and produces a LOW output when an object is detected. The object remains in front of the sensor for 80 ms, but the detection must be accepted only if the signal is LOW continuously for at least 50 ms. Which implementation best satisfies this requirement without blocking other time-critical tasks? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Record millis() at the first LOW and accept when LOW persists for 50 ms
(b) Count exactly 50 loop iterations after the first LOW reading
(c) Use analogRead() and accept when the value exceeds 50
(d) Use delay(50) after every LOW reading

Q42. An IR receiver module connected to an Arduino Uno gives an unstable digital output when a motor starts. The sensor and motor use the same 5 V supply. Which hardware change most directly reduces supply-induced false detections? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Connect the sensor ground through the motor winding
(b) Replace the digital input with an unused PWM output
(c) Increase the sensor threshold by changing the Arduino clock speed
(d) Add local decoupling near the sensor and separate motor return paths

Q43. A reflective IR sensor is calibrated using a white target, but the system later detects a black target at the same distance inconsistently. What is the most appropriate design response? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Disable ambient-light compensation to make the output digital
(b) Characterize target reflectivity and calibrate the detection threshold accordingly
(c) Use a fixed threshold selected only from the white-target reading
(d) Increase the Arduino supply voltage above 5 V

Q44. A sensor output is connected to Arduino Uno pin D2, and an interrupt is configured for CHANGE . A moving object causes 12 transitions within 3 ms because of surface texture. Which

strategy best prevents one object from being counted multiple times? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Replace CHANGE with LOW and keep the same counting logic
(b) Increment the count on every interrupt transition
(c) Disable interrupts permanently after the first transition
(d) Use edge qualification and ignore further events during a timed lockout

Q45. An Arduino Uno reads an analog IR receiver on A0. With the sensor disconnected, the reading changes randomly between 0 and 1023. Which conclusion is most defensible? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) The ADC is automatically generating a valid distance measurement
(b) The floating input is picking up noise and needs a defined electrical state
(c) The Arduino reference voltage has become exactly zero
(d) The IR emitter is operating at an excessive duty cycle

Q46. A 5 V IR sensor module has an output that may reach 5 V, while the receiving controller input is a 3.3 V-only device. Which interface is the safest general solution? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Use a resistor divider or a proper 5 V-to-3.3 V level shifter
(b) Configure the receiving pin as an analog output
(c) Add a capacitor in series and leave the input otherwise unchanged
(d) Connect the output directly because logic inputs always tolerate 5 V

Q47. An Arduino Uno detection system must operate for several hours from a battery. The IR emitter is continuously enabled even when no measurement is being made. Which modification most effectively reduces average sensor power while preserving reliable detection? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Increase the ADC resolution by changing analogRead() syntax
(b) Pulse the IR emitter and sample synchronously when detection is needed
(c) Set the sensor input pin to OUTPUT-HIGH continuously
(d) Increase serial monitor output frequency

Q48. Two IR detection sensors are installed 8 cm apart, but each emitter spreads radiation widely and both receivers respond to the same object. Which change most directly improves spatial discrimination? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Replace both sensor grounds with separate floating grounds
(b) Read both inputs only after calling delay(1000)
(c) Increase the baud rate of the serial interface
(d) Use optical shielding and narrower emitter-receiver alignment

Q49. An IR sensor output is active LOW. The Arduino code uses pinMode(sensorPin, INPUT_PULLUP) , but the sensor output is a push-pull output that drives HIGH during detection. What is the primary concern? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) The ADC loses all resolution on digital pins
(b) The sensor output and pull-up necessarily create a damaging short circuit
(c) The software polarity and assumed idle state may be interpreted incorrectly
(d) The pull-up changes the Arduino clock frequency

Q50. A detection routine stores the current sensor state in a variable shared with an interrupt service routine. Which declaration is required when the variable can change asynchronously and is read in the main program? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) volatile bool sensorState
(b) static double sensorState
(c) const int sensorState
(d) register char sensorState

Q51. An Arduino Uno measures the time between IR sensor edges using micros() . The measurement occasionally becomes incorrect after the timer counter wraps around. Which expression is wraparound-safe for unsigned timestamps now and previous ? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) if (now > previous + interval)
(b) if (previous - now >= interval)
(c) if (now - previous >= interval)
(d) if (now + interval > previous)

Q52. An analog IR sensor produces 2.60 V, and the Arduino Uno uses its default 5.00 V analog reference. What ADC value is theoretically expected for the 10-bit converter? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) About 133
(b) About 512
(c) About 858
(d) About 532

Q53. A sensor's analog output has a source impedance of 200 k Ω . Consecutive Arduino Uno ADC readings differ significantly, especially after switching from another channel. What is the best firmware-level correction? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Discard the first conversion after channel selection and allow settling
(b) Set all unused digital pins to INPUT without pull-ups
(c) Drive the analog pin HIGH before every conversion
(d) Take one reading only and increase the serial baud rate

Q54. A binary IR sensor signal contains short noise pulses. The system must detect a genuine object only when at least 4 of the latest 5 samples are active. Which method implements this rule most directly? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Average the samples as floating-point voltages only
(b) Trigger whenever any two nonconsecutive samples are active
(c) Accept the first active sample and reset the processor
(d) Use a running five-sample window and count active samples

Q55. An IR sensor is connected to an Arduino Uno input located near a PWM motor-control wire. The detection fails only when the motor duty cycle changes rapidly. Which investigation should be performed first? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Check signal integrity, grounding, decoupling, and wire routing
(b) Increase the number of LCD characters displayed
(c) Replace digitalRead() with Serial.read()
(d) Change the C++ variable names to shorter identifiers

Q56. A sensor output is normally HIGH, but it becomes disconnected intermittently. The Arduino input then registers random transitions. Which configuration is most appropriate if the sensor interface is an open-collector output that asserts LOW? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Use OUTPUT_LOW and read the same pin
(b) Use INPUT_PULLDOWN and treat HIGH as the active state
(c) Use INPUT_PULLUP and treat LOW as the active state
(d) Use a floating INPUT and average digital readings

Q57. An Arduino Uno drives an IR LED directly from a digital pin through a 100 Ω resistor from a 5 V supply. The LED forward voltage is 1.2 V. Ignoring pin limits, the nominal current is approximately what value? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) 50 mA

(b) 38 mA
(c) 12 mA
(d) 24 mA

Q58. A reflective IR detector must distinguish sunlight from a nearby modulated IR emitter. Which receiver strategy gives the strongest rejection of steady ambient illumination? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Measure only the total unmodulated photodiode current
(b) Modulate the emitter and synchronously detect its modulation
(c) Increase the receiver gain until the ADC saturates
(d) Use a longer cable with no reference conductor

Q59. A detection output drives a relay module from Arduino Uno pin D8. The relay resets the Arduino when energized. Which correction is most appropriate for a bare relay coil driven by a transistor? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) Connect the diode in series with the Arduino input pin
(b) Use a capacitor across the sensor output instead of coil suppression
(c) Remove the transistor so the pin absorbs the inductive transient
(d) Place a flyback diode across the coil and provide adequate supply decoupling

Q60. A sensor is sampled every 10 ms, and a detection requires 5 consecutive active samples. An object enters the sensing region immediately after a sample. What is the worst-case confirmation latency, excluding sensor response time? [Tag: Design and implementation of an Arduino Uno and IR sensor-based detection system | Hard]

(a) 40 ms
(b) 50 ms
(c) 10 ms
(d) 60 ms

6.2 Section B: Comprehensive Subjective Problem with Detailed Analytical Rubric

Problem 6 **[Workshop Practical / Viva Scenario — 5/10 Marks]**

1 An automated obstacle detection and count-alert system is built using an Arduino Uno (ATmega328P) and an active-LOW Infrared (IR) Proximity Sensor module. (a) Provide the complete circuit schematic wiring interface between the IR sensor, Arduino Uno, and an indicator buzzer/LED. (b) Write a robust Arduino C++ sketch incorporating software debounce to count detected objects and activate an alert when more than 50 objects pass.

Step-by-Step Analytical / Diagnostic Solution:

(a) Hardware Interface Wiring:
- **IR Sensor Module**:
* VCC → Arduino 5V
* GND → Arduino GND
* OUT → Arduino Digital Pin 2 (External Interrupt INT0)
- **Indicator LED**: Anode to Digital Pin 13 through a 220 Ω series current-limiting resistor; Cathode to GND.
- **Piezo Buzzer**: Positive terminal to Digital Pin 8; Negative terminal to GND.

(b) Embedded Firmware Implementation:

```
const int irSensorPin = 2;
const int ledPin = 13;
const int buzzerPin = 8;

volatile unsigned long objectCount = 0;
volatile unsigned long lastTriggerTime = 0;
const unsigned long debounceDelay = 150; // milliseconds

void setup() {
  pinMode(irSensorPin, INPUT_PULLUP);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
  Serial.println(\"--- ECE120: Arduino IR Object Counter Active ---\");
}

void loop() {
  int sensorState = digitalRead(irSensorPin);
  if (sensorState == LOW) { // Active LOW detection
    unsigned long currentTime = millis();
    if (currentTime - lastTriggerTime > debounceDelay) {
      objectCount++;
      lastTriggerTime = currentTime;
      digitalWrite(ledPin, HIGH);
      Serial.print(\"Object Detected! Cumulative Count: \");
      Serial.println(objectCount);
      if (objectCount >= 50) {
        digitalWrite(buzzerPin, HIGH); // Alarm threshold
      }
      delay(50);
      digitalWrite(ledPin, LOW);
    }
```

```
  }
}
```

Examiner Rubric (10 Marks):
- Interfacing schematic & pin mapping with current limiting: 3 Marks
- Complete setup() and pin configuration logic: 2 Marks
- Software debounce algorithm using non-blocking millis(): 3 Marks
- Threshold counter condition and actuator activation: 2 Marks

6.3 Section C: Laboratory Viva-Voce Questions with Authoritative Solutions

Viva Question: Explain the working principle of the LM393 comparator inside an IR proximity sensor module.
Technical Answer: The IR sensor features an IR LED emitter and photodiode receiver. Reflected IR light from an obstacle creates a reverse photocurrent, developing a proportional voltage at the non-inverting input ($IN+$) of the LM393. The inverting input ($IN-$) is held at a reference voltage set by the multi-turn potentiometer. When reflected voltage exceeds reference, the comparator output transitions from HIGH to LOW, signaling an obstacle.

Viva Question: What is the resolution of the Arduino Uno Analog-to-Digital Converter (ADC)?
Technical Answer: The ATmega328P on the Arduino Uno includes a 10-bit successive approximation ADC. With default reference voltage $V_{\text{ref}} = 5.0$ V, it quantizes input voltages into $2^{10} = 1024$ discrete steps (0 to 1023). The voltage resolution per LSB is $5.0\text{ V}/1024 \approx 4.8828$ mV.

Viva Question: What are the limitations of optical IR proximity sensors in industrial environments?
Technical Answer: IR sensors suffer from: (1) Ambient sunlight or incandescent infrared interference that saturates the photodiode, (2) Surface reflectivity dependency (dark/matte objects absorb IR and fail to trigger, while white/glossy surfaces trigger at long distances), and (3) Lens dust/oil contamination in harsh factory atmospheres.

6.4 Section D: Workshop Troubleshooting & Fault Diagnostic Case Study

Troubleshooting Scenario: Arduino Resets Every Time Buzzer/Relay Activates
When an Arduino obstacle detection circuit triggers an external 5V electromagnetic relay or buzzer, the Arduino Uno immediately resets and re-runs ‘setup()‘.

Diagnostic Cause: Inductive kickback and power rail brownout. Inductive actuators draw heavy initial surge current and produce high-voltage negative flyback spikes when switched off ($V = L\frac{di}{dt}$), pulling the +5V rail below the ATmega328P brown-out detection (BOD) threshold of 4.3V.
Hardware Fix: (1) Place a reverse-biased flyback diode (1N4007) directly across the relay coil terminals. (2) Power inductive actuators from a separate external power supply, keeping ground rails common. (3) Install a 100 μF electrolytic capacitor in parallel with a 0.1 μF ceramic capacitor across the Arduino power rails for supply decoupling.